

National Audit of Pulmonary Hypertension Great Britain, 2018-19

Great Britain

24 October 2019

Tenth Annual Report

Introduction: Foreword

It is a great pleasure to introduce the 10th National Audit of Pulmonary Hypertension (NAPH) annual report which clearly shows how the Audit is driving clinical improvement across UK Specialist Pulmonary Hypertension (PH) Centres. This has been achieved through the enthusiasm and thorough engagement of clinical staff in the Audit.



Ten years ago the Audit started to collect and report upon prospective data, and track clinical outcomes and processes with the aim of improving patient care and informing future service planning. This year, for example, we see shortening of the time it takes to see new patients and make their diagnosis promptly. The striking expansion of the patient population year on year has been important in planning for the future of the PH centres. Unusually after 10 years it shows no signs of levelling off.

The success of this Audit is attributable to a great many people and organisations. PHA UK funded its creation, have provided much help and useful advice, and now produce a patient-friendly version of the annual report. The Commissioners picked up the funding tab from PHA UK and have wholeheartedly supported the Audit ever since. The healthcare professionals, administrative staff and IT staff at the PH centres have undertaken an enormous workload without additional funding to provide the best quality data. The PH Outcomes Group have developed and guided the National Standards used in the report and much else besides. The team at NHS Digital have overseen information governance while storing and analysing the data.

It has been my privilege to lead the Audit since its inception. I believe that it is now time to hand over the reins of this unique dataset to its new clinical lead, David Kiely, to drive it forward into its second decade.

Simon Gibbs, Lead Clinician, National Audit of Pulmonary Hypertension

Introduction: Overview

- The National Audit of Pulmonary Hypertension (NAPH) sets out to measure the quality of care provided to people referred to pulmonary hypertension services in Great Britain.
- Data is collected and submitted by hospital staff at pulmonary hypertension services in Great Britain. Efforts continue to be made to incorporate Northern Ireland into the Audit again.
- The Audit is delivered by NHS Digital and commissioned by NHS England and supported by NHS Scotland, NHS Wales (GIG Cymru), the Pulmonary Hypertension Association (PHA UK) and the National Pulmonary Hypertension Centres of United Kingdom and Ireland Physicians' Committee.
- Please see the [Explanatory Notes](#) and accompanying Data Quality Statement for information on data quality.



England



Introduction: Audit questions

Pulmonary hypertension (PH) is raised blood pressure within the pulmonary arteries, which are the blood vessels that supply the lungs. It is a serious medical condition that can damage the right side of the heart, making the heart less efficient at pumping blood around the body and getting oxygen to the muscles.

The National Audit of Pulmonary Hypertension measures the quality of care provided to people referred to PH services in England (including patients referred from Wales) and Scotland, by answering a number of questions:

- Are pulmonary hypertension services appropriate?
- Are patients receiving the right treatment in a timely manner?
- What are the outcomes for patients with pulmonary hypertension?



The Audit answers these questions through the measurement of a number of professionally agreed **standards** reported at a national and centre level.

This report is of particular interest to commissioners and healthcare providers to monitor the quality and effectiveness of PH services. It may also be of interest to researchers and charities working on this subject and to members of the public, including those with pulmonary hypertension.

Introduction: Terminology



Abbreviations used in the report

APAH	associated pulmonary arterial hypertension
CTD	connective tissue disease
CTEPH	chronic thromboembolic PH
IPAH	idiopathic, heritable or drug-induced PAH
NAPH	National Audit of Pulmonary Hypertension
PAH	pulmonary arterial hypertension
PDE5	phosphodiesterase 5
PEA	pulmonary endarterectomy
PH	pulmonary hypertension
PHA UK	Pulmonary Hypertension Association UK
WHO	World Health Organization

Patient groups referred to in the report

- **Managed patients** have a referral active at any point during the appropriate year (10,572 in 2018-19).
- **Active patients** have a referral active on the 31 March of the appropriate year (8,351 in 2019) .
- **Active drug therapies** represent patients with at least one pulmonary hypertension drug therapy being prescribed on the 31 March of the appropriate year (4,683 in 2019).
- The **cohort for longitudinal analysis** are adult patients whose first referral was on or after 1 April 2009 (23,311 in 2009-19).

Specialist PH centre names used in the report

Location	PH centre (short name)
Glasgow	Golden Jubilee
London	Great Ormond Street
London	Imperial College
Newcastle	Newcastle
London	Royal Brompton and Harefield
London	Royal Free
Cambridgeshire	Royal Papworth
Sheffield	Sheffield

For further information, see:

[Glossary: Acronyms and definitions](#)

[Glossary: Patient cohorts](#)

[Glossary: Specialist PH centres.](#)

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The following documents are published alongside the main report

- Supplementary Survival Analysis: Sex, Age and WHO Functional Class
- Tables and Charts (Excel)
- Open Data (csv)
- Data Quality Report
- Dana Point Classification
- Staff Involved in Audit

National Audit of Pulmonary Hypertension

Key messages:

Results and Findings



Key messages: Successes



At national level, 11 of the 13 National Standards were met.

This provides reassurance that PH centres are delivering high standards of care for the majority of audit measures.



All 8 specialist PH centres covering England, Scotland and Wales took part in the 2018-19 Audit:

All specialist PH centres are expected to participate in the NAPH.

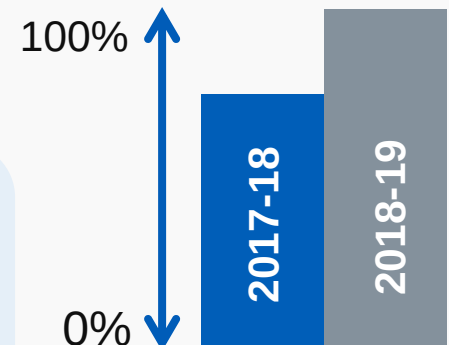
This year, for the first time, all PH centres met the following National Standard:

- 'New patients should be seen or discharged within 30 days'

"At **Royal Brompton and Harefield**, we analyse our data against the standards every two months and identify where we need to improve. In addition, our data entry manager continuously looks for errors and missing data. It is a real team effort."

John Wort, Consultant Physician

New patients seen or discharged within 30 days



Key messages: Work in progress



Adherence to National Standards varies between PH centre

Every PH centre misses at least one National Standard. Where standards are not met, patients may experience lower standards of care than in higher-performing organisations.

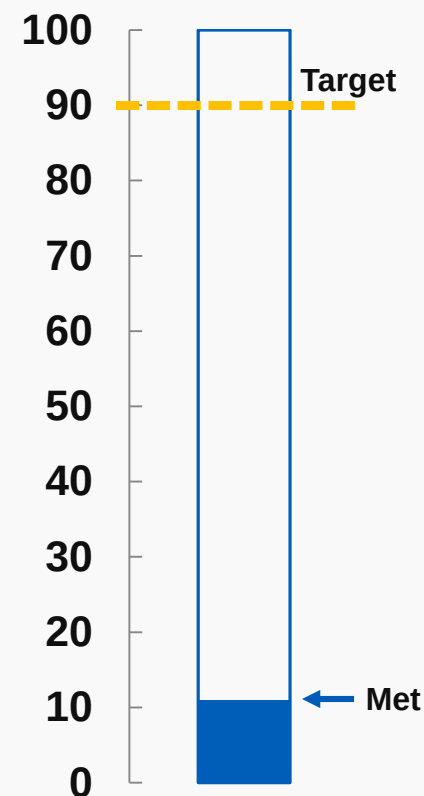
At both national and PH centre level, the proportion of cases meeting National Standard 13 (11 per cent) falls far below the target level (90 per cent):

‘Waiting times for pulmonary endarterectomy should be <4 months’.

The median time from diagnosis to PEA surgery has improved since 2016 but unfortunately the majority of patients still do not meet standard 13. There are multifactorial reasons but the commissioners and PH centres continue to work together through pathway meetings and planning for increasing surgical capacity at Royal Papworth with the aim of meeting this challenging standard.

Kathy Blacker, *Lead Commissioner, NHS England*

Cases meeting NS 13



National Audit of Pulmonary Hypertension

National Standards: Summary



National Standards: Overview

Methodology

This is the **fourth time** that the Audit has used standards against which to measure clinical practice, and the third time that results have been published at specialist PH centre level.

Measurement of clinical practice against professional clinical standards is a key principle of clinical audit. Once comparison to standards has taken place a clinical team can then develop local action plans for any changes that have been identified as necessary in order to improve their PH service.

The standards in the Audit were agreed for the eighth annual report. Targets and standards have been amended and new standards introduced by the Pulmonary Hypertension Outcomes Group (PHOG).

Standards are applicable to all specialist PH centres, although they do not provide identical services. Thresholds for standards are not set to 100 per cent because the targets do not take every factor into account. Care may be given that is not in line with the standard without indicating it was incorrect or poor practice. Specialist PH centres should consider whether this is the case when reflecting on their results.



Meeting the standards

Proportions meeting each standard are **rounded** up or down to the nearest whole number. The standard is met if the rounded number is greater than or equal to the target value.



National Standards: Comparability

[Changes](#)

As in the previous NAPH report, results from previous years have been updated using:

1) the latest available information in the NAPH

All results in this report are derived from data extracted on 24 June 2019, meaning that results for earlier audit years (2015-16 to 2017-18) will sometimes differ from those published in previous reports.

[Why has a new extract been used?](#) It is important to take the latest and most accurate cut of the NAPH data. NAPH is a continuous collection and PH centres are able to add, amend and delete information about patients used in earlier cohorts. This is particularly true for the 10th Annual Report, after PH centres reviewed the diagnosis of all IPAH patients in the NAPH during early 2019 (see [Slide 73](#)).

2) the latest methodology

Methodology is constantly reviewed and revised if required. Details of methodological changes to the National Standards are on the next [Slide](#). The updated methodology has been applied to all audit years in the report.

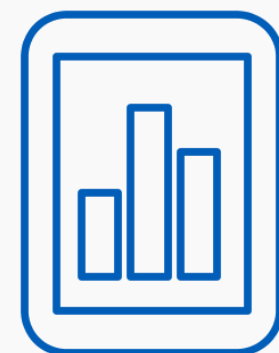
[Why have earlier years been updated with the new methodology?](#) To maintain the clarity of this report, it is important that the same methodology is applied across all audit years in the publication.

3) the latest targets

Target levels are constantly reviewed and revised if required. Changes to the targets for the 10th Annual Report are outlined on [Slide 13](#), with updated targets applied to all audit years in the report.

[Why have earlier years been updated with the new targets?](#) To enhance the clarity of this report, it is important that the targets shown in the 10th Annual Report tables are consistent across all audit years.

Following the above changes, the figures produced for the 10th Annual Report can be considered to supersede those previously published.



National Standards: Updates

Changes

Updated numbering: To enhance the readability of this report, Standard numbers have been updated to match the order of their appearance in this publication.

Updated targets:

To reflect the national increase in the number of patients managed by the specialist PH centres, the target number for National Standard 2 has been raised from 250 to 300 at adult PH centres and from 50 to 100 at the children's PH centre.

To take account of more virtual consultations for sick patients unable to travel to their designated PH centre, the target for National Standard 12 has been reduced from 95 to 90 per cent.

Updated methodology:

National Standard 4 'New patients should be seen or discharged within 30 days' at the children's centre has been restricted to cover PAH/CTEPH patients only. Previously all patients were included. This standard has been altered to focus on children with PAH and CTEPH because these patients may require urgent treatment which is only available at a specialist PH centre.

To account for patients who need to be stabilized with prostacyclin before undergoing cardiac catheterization, National Standard 5 'Patients receiving a PH drug should have pre-treatment cardiac catheterization' has been extended to include catheterization within 2 weeks after the start of the first PH drug.

National Standard 7 'Patients should have a pre-treatment vasoreactivity study recorded' is new this year and is reported at national level only. The standard will be reported at PH centre level in subsequent reports.

Further information on the targets and methodological changes is on [Slide 82](#).



National Standards: Summary by audit year

Table NSS 1: National Standards: Summary by audit year, Great Britain, 2015-19

[Methodology](#)

No.	National Standard Description	Target	Audit Year				Trend ¹ 17-18 to 18-19	
			2015-16 Per cent	2016-17 Per cent	2017-18 Per cent	2018-19 Per cent	p≤0.05	
1	PH centres should participate in the Audit ♦	All	7 of 8	8 of 8	8 of 8	8 of 8	N/A	
2	PH centres should see a sufficient number of patients \$	All	7 of 7	8 of 8	8 of 8	8 of 8	N/A	
3	Patients should have a timely diagnosis ♦	95	96	98	98	99	Up	▲
4	New patients should be seen or discharged within 30 days \$	50	47	53	60	63	Up	▲
5	Patients receiving a PH drug should have pre-treatment cardiac catheterization ♦ \$	95	91	96	96	97	Static	—
6	Patients should have pre-treatment WHO functional class and an exercise test recorded ♦	90	80	85	98	97	Static	—
7	Patients should have a pre-treatment vasoreactivity study recorded ♦	80				57	N/A	
8	New patients should begin drug therapy within 12 weeks of referral	80	76	82	81	81	Static	—
9	Patients receiving a PH drug should have had a PH diagnosis recorded	99	100	100	100	100	Static	—
10	First line drug therapy for PAH should be a phosphodiesterase 5 (PDE5) inhibitor	80	90	92	93	91	Static	—
11	Patient quality of life should be recorded ♦	90	78	89	96	96	Static	—
12	Patients receiving a PH drug should have an annual consultation ♦ \$	90	96	96	96	96	Static	—
13	Waiting times for pulmonary endarterectomy should be <4 months	90	14	13	18	11	Static	—

Key: Target n Target met n Target not met n Not applicable N/A p≤0.05 See [Glossary](#)

▲ = Upwards trend
— = Static trend
▼ = Downward trend



♦ = Identified by PHA UK in 2016 as the 6 most important NAPH National Standards.
♦ = New National Standard. \$ = Change to target/definition (see [Slide 83](#)).
N/A = Not applicable. 1. Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend.

National Standards: Summary by PH centre

Table NSS 2: National Standards: Summary by PH centre, Great Britain, 2018-19

[Methodology](#)

National Standard		Target	PH centre (per cent)								
No.	Description		National total	Golden Jubilee	Great Ormond Street	Imperial College	Newcastle	Royal Brompton and Harefield	Royal Free	Royal Papworth	Sheffield
1	PH centres should participate in the Audit ♦	All	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
2	PH centres should see a sufficient number of patients \$	All	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3	Patients should have a timely diagnosis ♦ \$	95	99	99	89	100	94	99	99	99	100
4	New patients should be seen or discharged within 30 days \$	50	63	74	100	62	78	57	75	53	55
5	Patients receiving a PH drug should have pre-treatment cardiac catheterization ♦ \$	95	97	99	N/A	92	100	97	97	97	98
6	Patients should have pre-treatment WHO functional class and an exercise test recorded ♦	90	97	100	N/A	92	94	96	97	94	98
7	Patients should have a pre-treatment vasoreactivity study recorded ♦	80	57		N/A						
8	New patients should begin drug therapy within 12 weeks of referral	80	81	61	N/A ¹	89	91	88	87	75	79
9	Patients receiving a PH drug should have had a PH diagnosis recorded	99	100	99	N/A	100	100	100	100	100	100
10	First line drug therapy for PAH should be a phosphodiesterase 5 (PDE5) inhibitor	80	91	89	N/A	100	86	84	95	96	91
11	Patient quality of life should be recorded ♦	90	96	94	N/A	95	100	93	98	93	99
12	Patients receiving a PH drug should have an annual consultation ♦ \$	90	96	97	95	95	99	97	97	96	94
13	Waiting times for pulmonary endarterectomy should be <4 months	90	11	23	N/A	10	0	0	0	23	6

♦ = Identified by PHA UK in 2016 as the 6 most important NAPH National Standards.

♦ = New National Standard. \$ = Change to target/definition (see [Slide 83](#)).

N/A = Not applicable. 1. No patients applicable.

Key: Target n Target not met n
Target met n Not applicable N/A

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NS 1: PH centres should participate in the Audit: Overview

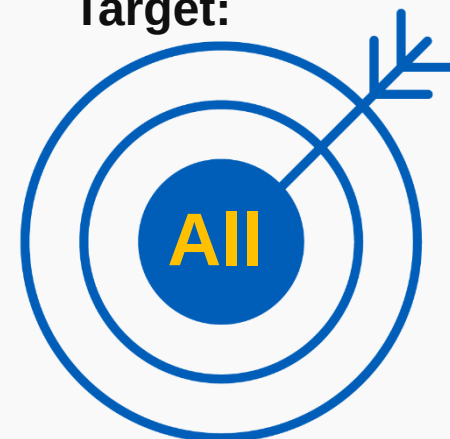
Audit standard

All specialist PH centres are expected to participate¹ in the National Audit of Pulmonary Hypertension².

Why is this important?

Participation in the Audit can be considered a quality measure of a hospital since it indicates a willingness to share data and change practice to improve clinical performance.

Target:



Key finding

All specialist PH centres participated in the Audit.



Target: All centres

Met at: 8 of 8 centres

Trend: Full participation for last 3 years

Notes: 1. PH centres will determine if they are participating. Failure to participate will remove them from all other standards.

2. The NHS England PH service specification [A11/S/a](#) 2013-14 states that "All designated PH centres will be required to submit data to the National Pulmonary Hypertension Service" (p.12) which is assumed to be a reference to the Audit.



NS 2: PH centres should see a sufficient number of patients: Overview

Audit standard

Adult specialist PH centres¹ are expected to manage at least 300 patients per annum with pulmonary arterial hypertension (PAH) and / or chronic thromboembolic pulmonary hypertension (CTEPH)^{2,3}.

The children's specialist PH centre¹ is expected to manage at least 300 patients per annum, of whom 100 patients per annum with PAH or CTEPH².

Why is this important?

Evidence from other disease areas suggests that best outcomes for patients are obtained in high volume PH centres. High volume PH centres also enable provision of a full range of investigation modalities, specialist nursing support and involvement of a wider multi-professional team with expertise in patients with PAH or CTEPH. The recommended number of patients seen is based on European guidelines which are used by the UK specialist PH centres.

New changes to National Standard 2

To reflect the national increase in the number of patients managed by the specialist PH centres, the target number for National Standard 2 has been raised from 250 to 300 at adult PH centres and from 50 to 100 at the children's PH centre.

Target: 300 | 300 (100)

Met at: 8 of 8 centres

Trend: All met for last three years



Notes: 1. Adults refers to patients under the care of adult PH centres. Children are under the care of Great Ormond Street.

2. An open referral at any point in the year will constitute 'manage' within this context. An open referral indicates patients attending a specialist PH centre who include existing patients as well as new referrals. The latest diagnosis will determine PAH or CTEPH. 3. This standard is derived from ESC/ERS guidelines with the number of patients adjusted to the NHS England PH service specification [A11/S/a](#) 2013-14.



NS 2: PH centres should see a sufficient number of patients: Results

Table NS 2: PH centres meeting National Standard 2

Great Britain, 2016-19

Adults' PH centre (Target = 300)	Managed patients with PAH or CTEPH		
	Audit year		
	2016-17	2017-18	2018-19
Golden Jubilee	427	457	474
Imperial College	768	773	770
Newcastle	319	334	352
Royal Brompton and Harefield	627	641	683
Royal Free	713	757	791
Royal Papworth	957	967	959
Sheffield	1,727	1,803	1,926

All centres saw a sufficient number of patients in 2017-18.



[Changes](#)

[Methodology](#)

Key:

Target met	n
Target not met	n

Children's PH centre (Target = 100 PAH / CTEPH, 300 managed)	Managed patients with PAH or CTEPH			Managed patients		
	Audit year			Audit year		
	2016-17	2017-18	2018-19	2016-17	2017-18	2018-19
Great Ormond Street	154	150	147	354	356	351

Notes: 1. Definition: Adult specialist PH centres are expected to manage at least 300 patients with PAH or CTEPH in a year.
2. Definition: The children's specialist PH centre is expected to manage at least 300 patients in a year, including 100 patients with PAH or CTEPH.



NS 3: Patients should have a timely diagnosis: Overview

Audit standard

95 per cent of patients whose referral letter was received over six months ago should have a diagnosis recorded¹.

“**Royal Papworth** have introduced flag system to remind staff to input the missing information into the database. As the result all diagnoses are inputted in timely manner.”

Joanna Pepke-Zaba
Consultant Physician



National: 99%

Target: 95%

Met at: 6 of 8 centres

Trend: Up

Why is this important?

It is understandable that most people who have symptoms want to know what is causing them and find a suitable treatment as soon as possible.

Symptoms may get worse while waiting for a diagnosis of pulmonary hypertension. Specialist PH centres need to ensure that they are seeing their new patients in a timely fashion.



NS 3: Patients should have a timely diagnosis: Results

Table NS 3: Proportion of cases meeting National Standard 3, Great Britain, 2016-19

PH centre (Target = 95%)	Audit year			Trend ² 17-18 to 18-19	
	2016-17	2017-18	2018-19		
	Per cent	Per cent	Per cent	$p \leq 0.05$	
Golden Jubilee	99	100	99	Static	—
Great Ormond Street	92	95	89	Static	—
Imperial College	100	99	100	Static	—
Newcastle	97	98	94	Static	—
Royal Brompton and Harefield	98	99	99	Static	—
Royal Free	98	97	99	Up	▲
Royal Papworth	92	92	99	Up	▲
Sheffield	99	100	100	Static	—
National total	98	98	99	Up	▲

[Methodology](#)



Key finding

99 per cent of new patients are diagnosed within 6 months

Key:

▲ = Upwards trend

— = Static trend

▼ = Downward trend

Target met n

Target not met n

$p \leq 0.05$ See [Glossary](#)

Notes: 1. Definition: 95 per cent of patients whose referral letter was received over six months ago should have a diagnosis recorded.
2. Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend.



NS 4: New patients should be seen or discharged within 30 days: Overview

Audit standard

50 per cent of patients should have attended a consultation (inpatient, day case or outpatient) or have been discharged within 30 days of receipt of their first referral¹.

Why is this important?

This standard measures how quickly patients are seen by specialist PH centres following a new referral. In some patients, the symptoms of PH may progress between referral and the first consultation in a PH centre so it is desirable that patients are seen promptly. Unlike National Standard 3, this standard explores the proportion of patients who have had at least a consultation within a fixed time period.



Nationally: 63%

Target: 50%

Met at: 8 of 8 centres

Trend: Up

New changes to National Standard 4

Assessment of National Standard 4 at the children's centre has been restricted to cover PAH/CTEPH patients only. Previously all patients were included.

“Since **Golden Jubilee** introduced additional slots for new patients last year, most new patients have been waiting less than 4 weeks.”

Martin Johnson, *Consultant Physician*

Notes: 1. The NHS England PH service specification [A11/S/a](#) 2013-14 states that “There will be minimal delay between referral to a designated PH centre and an outpatient consultant appointment. All patients referred urgently will be able to see a specialist within one month of the PH centre receiving their referral and those who are severely symptomatic will be seen within 2 weeks.” (p.4). The Audit does not distinguish elective and urgent referrals so this proportion is arbitrary.



NS 4: New patients should be seen or discharged within 30 days: Results

Table NS 4: Proportion of cases meeting National Standard 4, Great Britain, 2016-19

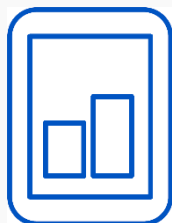
PH centre (Target = 50%)	Audit year			Trend ² 17-18 to 18-19	
	2016-17	2017-18	2018-19		
	Per cent	Per cent	Per cent	$p \leq 0.05$	
Golden Jubilee	17	42	74	Up	▲
Great Ormond Street ³	41	94	100*	Static	—
Imperial College	71	63	62	Static	—
Newcastle	19	38	78	Up	▲
Royal Brompton and Harefield	58	60	57	Static	—
Royal Free	55	71	75	Static	—
Royal Papworth	43	52	53	Static	—
Sheffield	64	63	55	Down	▼
National total	53	60	63	Up	▲

[Changes](#)

[Methodology](#)

Impact:

Newcastle **up** from 38 to 78 per cent



All met in 2018-19



Key:

- ▲ = Upwards trend
- = Static trend
- ▼ = Downward trend

Target met n

Target not met n

$p \leq 0.05$ See [Glossary](#)

Notes: * = Fewer than 10 cases in the denominator. 1. Definition: 50 per cent of patients should have attended a consultation (inpatient, day case or outpatient) or have been discharged within 30 days of receipt of their first referral. Assessment of National Standard 4 at the children's centre has been restricted to cover PAH/CTEPH patients only. Previously all patients were included. 2. Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend.



NS 5: Patients receiving a PH drug should have pre-treatment cardiac catheterization: Overview

Audit standard

95 per cent of adult patients receiving a pulmonary hypertension drug therapy (excluding patients diagnosed as pulmonary arterial hypertension associated with congenital heart disease¹) should have a cardiac catheterization recorded, either before or up to two weeks after treatment begins.

Why is this important?

Cardiac catheterization² is considered the best method for confirming the presence, nature and severity of pulmonary hypertension.

It is reasonable to be certain about the diagnosis before starting treatment.



Nationally: 97%

Target: 95%

Met at: 6 of 7 centres

Trend: Similar to previous audit year

New changes to National Standard 5

To account for patients who need to be stabilized with prostacyclin (epoprostenol, iloprost, treprostinil, etc.) before undergoing cardiac catheterization, National Standard 5 has been extended to include catheterization within 2 weeks after the start of the first PH drug.



NS 5: Patients receiving a PH drug should have pre-treatment cardiac catheterization: Results

Table NS 5: Proportion of cases meeting National Standard 5, Great Britain, 2016-19

Adults' PH centre (Target = 95%)	Audit year			Trend ² 17-18 to 18-19		Changes
	2016-17	2017-18	2018-19			Methodology
	Per cent	Per cent	Per cent	$p \leq 0.05$		DQ note
Golden Jubilee	100	99	99	Static	—	
Imperial College	100	98	92	Static	—	
Newcastle	89	98	100	Static	—	
Royal Brompton and Harefield	95	94	97	Static	—	
Royal Free	100	97	97	Static	—	
Royal Papworth	95	97	97	Static	—	
Sheffield	94	95	98	Up	▲	
National total	96	96	97	Static	—	



Key findings

All but one PH centres meet the standard.

Key:

- ▲ = Upwards trend
- = Static trend
- ▼ = Downward trend

Target met n

Target not met n

$p \leq 0.05$ See [Glossary](#)

Notes: 1. Definition: 95 per cent of adult patients receiving a pulmonary hypertension drug therapy (excluding patients diagnosed as pulmonary arterial hypertension associated with congenital heart disease) should have a cardiac catheterization recorded before or up to two weeks after treatment begins. 2. Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend.



NS 6: Patients should have pre-treatment exercise test and WHO functional class recorded: Overview

Audit question

Ninety per cent of adult patients with PAH / CTEPH should have a WHO functional class¹ and exercise test² (six-minute walk or incremental shuttle) recorded before they begin any pulmonary hypertension drug therapy.

Why is this important?

The World Health Organization (WHO) functional class¹ is a measure of the extent of functional limitation due to pulmonary hypertension and is a prognostic factor. Patients in different WHO functional classes require different PAH drugs and combinations of these drugs.

The extent of exercise limitation due to PAH / CTEPH can be assessed quantitatively by a formal exercise test. Both WHO functional class and exercise performance may improve with treatment and by recording this information a response to treatment can be documented later.



Nationally: 97%

Target: 90%

Met at: 7 of 7 centres

Trend: All met for 2nd consecutive year

“Through a multi-disciplinary team consultation, pre-treatment exercise test and WHO functional class have been incorporated as standard into the new patient assessment process at **Newcastle**.”

Alan Greenhalgh, Research Nurse³



NS 6: Patients should have pre-treatment exercise test and WHO functional class recorded: Results

Table NS 6: Proportion of cases meeting National Standard 6, Great Britain, 2016-19

Adults' PH centre (Target = 90%)	Audit year			Trend ² 17-18 to 18-19	
	2016-17	2017-18	2018-19		
	Per cent	Per cent	Per cent	$p \leq 0.05$	
Golden Jubilee	96	99	100	Static	—
Imperial College	95	100	92	Down	▼
Newcastle	87	98	94	Static	—
Royal Brompton and Harefield	96	93	96	Static	—
Royal Free	98	99	97	Static	—
Royal Papworth	91	96	94	Static	—
Sheffield	73	98	98	Static	—
National total	85	98	97	Static	—

[Methodology](#)

All met in
2018-19



Key:

- ▲ = Upwards trend
- = Static trend
- ▼ = Downward trend

Target met n

Target not met n

$p \leq 0.05$ See [Glossary](#)

Notes: 1. Definition: 90 per cent of adult patients with PAH / CTEPH should have a WHO functional class and an exercise test (six-minute walk or incremental shuttle) recorded before they begin any pulmonary hypertension drug therapy. 2. Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend.



NS 7: Patients should have a pre-treatment vasoreactivity study recorded: Overview

Audit standard

Eighty per cent of adult patients who have idiopathic, heritable or drug-induced pulmonary arterial hypertension (IPAH) should have undergone a vasoreactivity study before starting PAH drug therapy.



National: 57%

Target: 80%

To be assessed at PH centre-level in the next report

[Methodology](#)

Why is this important?

The information provided by undergoing a vasoreactivity study determines whether the patient with IPAH is most likely to respond to treatment with calcium channel blockers in preference to other drug therapies including prostanoids, endothelin antagonists or PDE5 inhibitors.

Introduction of National Standard 7

This is the first year that this National Standard has been reported. In future years the standard will also be reported at specialist centre level.



NS 8: New patients should begin drug therapy within 12 weeks of referral: Overview

Audit standard

Eighty per cent of new referrals who have a diagnosis of idiopathic, heritable, anorexigen-induced or connective tissue disease associated PAH and are in WHO functional class II, III or IV, should commence PAH drug therapy within 12 weeks of receipt of their first referral^{1,2}.



National: 81%

Target: 80%

Met at: 4 of 7 centres

Trend: One fewer adult PH centre meeting target

“At **Newcastle**, improvement work on the 30-day target has also resulted in the ability to treat patients sooner, within the 12-week window.”

Alan Greenhalgh
Research Nurse

Why is this important?

For patients with some types of PAH it is important to start drug treatment in a timely manner to try to avoid deterioration as a result of progressive disease. This measurement of the time it takes from referral to treatment provides an overview of the timeliness of clinical processes.



NS 8: New patients should begin drug therapy within 12 weeks of referral: Results

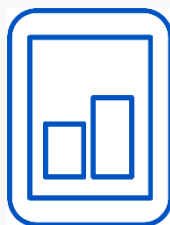
Table NS 8: Proportion of cases meeting National Standard 8, Great Britain, 2016-19

PH centre (Target = 80%)	Audit year			Trend ² 17-18 to 18-19	
	2016-17	2017-18	2018-19		
	Per cent	Per cent	Per cent	<i>p</i> ≤0.05	
Golden Jubilee	48	58	61	Static	—
Great Ormond Street	100*	100*	N/A ³	N/A	
Imperial College	100	83	89	Static	—
Newcastle	82	33	91	Up	▲
Royal Brompton and Harefield	80	87	88	Static	—
Royal Free	82	94	87	Static	—
Royal Papworth	68	93	75	Static	—
Sheffield	87	86	79	Static	—
National total	82	81	81	Static	—

[Methodology](#)

Impact

Newcastle **up** from 33 to 91 per cent



Key:

Target met n

Target not met n

p≤0.05 See [Glossary](#)

▲ = Upwards trend

— = Static trend

▼ = Downward trend

Notes: * = Fewer than 10 cases in the denominator. **1.** Definition: 80 per cent of new referrals who have a diagnosis of idiopathic, heritable, anorexigen-induced or connective tissue disease associated PAH and are in WHO functional class II, III or IV, should commence PAH drug therapy within 12 weeks of receipt of their first referral. **2.** Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend. **3.** No patients applicable.



NS 9: Patients receiving a PH drug should have had a PH diagnosis recorded: Overview

Audit standard

Ninety-nine per cent of adult patients receiving a pulmonary hypertension drug therapy should have a pulmonary hypertension diagnosis recorded.



National: 100%

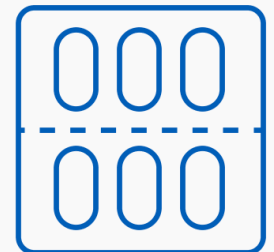
Target: 99%

Met at: 7 of 7 centres

Trend: All met for last three years

Why is this important?

Specialist pulmonary hypertension drugs should only be prescribed for selected patients with pulmonary hypertension. It is important to confirm that a patient, for whom a drug is prescribed, has a qualifying diagnosis.



NS 9: Patients receiving a PH drug should have had a PH diagnosis recorded: Results

Table NS 9: Proportion of cases meeting National Standard 9, Great Britain, 2016-19

Adults' PH centre (Target = 99%)	Audit year			Trend ² 17-18 to 18-19	
	2016-17	2017-18	2018-19	<i>p</i> ≤0.05	
	Per cent	Per cent	Per cent		
Golden Jubilee	99	99	99	Static	—
Imperial College	100	100	100	Static	—
Newcastle	100	100	100	Static	—
Royal Brompton and Harefield	100	99	100	Static	—
Royal Free	100	100	100	Static	—
Royal Papworth	99	100	100	Static	—
Sheffield	100	100	100	Static	—
National total	100	100	100	Static	—

[Methodology](#)



Key finding

Almost 100 per cent of patients receiving a PH drug have a PH diagnosis

Key:

- ▲ = Upwards trend
- = Static trend
- ▼ = Downward trend

Target met n

Target not met n

p≤0.05 See [Glossary](#)

Notes: 1. Definition: 99 per cent of adult patients receiving a pulmonary hypertension drug therapy should have a pulmonary hypertension diagnosis recorded. 2. Variation between audit years can occur in a 'static' trend. Where there are fewer cases are involved, a greater degree of variation is possible within a 'static' trend.



NS 10: First line drug therapy for PAH should be a PDE5 inhibitor: Overview

Audit standard

First line drug therapy for pulmonary arterial hypertension should be a phosphodiesterase 5 (PDE5) inhibitor in 80 per cent or more of adult patients, where calcium channel blockers are not the first drugs given.



National: 91%

Target: 80%

Met at: 7 of 7 centres

Trend: All met for last three years

Why is this important?

Pulmonary arterial hypertension can be treated with three different groups of medicines. The NHS England policy¹ requires that a PDE5 inhibitor² (sildenafil or tadalafil) is used first on economic grounds.

Nevertheless, this drug is contraindicated³ in some patients. According to NHS England policy, the patients who manifest the most severe symptoms (in WHO functional class IV) may be recommended a prostacyclin as first line therapy instead.

Notes: 1. The NHS England Clinical Commissioning Policy [A11/P/b](#) 2014 states that “Monotherapy with an oral PDE5I will be routinely commissioned as first line therapy.” (p.12).

2. Phosphodiesterase 5 (PDE5) inhibitors inhibit the enzyme PDE5, leading to relaxation of the blood vessels and increased blood flow.

3. Medical advice that suggests a particular drug or treatment should not be used in some circumstances.



NS 10: First line drug therapy for PAH should be a PDE5 inhibitor: Results

Table NS 10: Proportion of cases meeting National Standard 10, Great Britain, 2016-19

Adults' PH centre (Target = 80%)	Audit year			Trend ² 17-18 to 18-19	
	2016-17	2017-18	2018-19		
	Per cent	Per cent	Per cent	<i>p</i> ≤0.05	
Golden Jubilee	90	87	89	Static	—
Imperial College	84	98	100	Static	—
Newcastle	93	97	86	Static	—
Royal Brompton and Harefield	91	91	84	Static	—
Royal Free	100	96	95	Static	—
Royal Papworth	90	85	96	Static	—
Sheffield	92	93	91	Static	—
National total	92	93	91	Static	—

[Methodology](#)



Key finding

All PH centres met the standard for the third consecutive year. Meeting this standard is partially dependent on the centre's casemix.

Key:

- ▲ = Upwards trend
- = Static trend
- ▼ = Downward trend

Target met n

Target not met n

p≤0.05 See [Glossary](#)

Notes: **1.** Definition: First line drug therapy for pulmonary arterial hypertension should be a phosphodiesterase 5 (PDE5) inhibitor in more than 80 per cent of adult patients, excluding those patients whose first drug therapy was a calcium channel blocker. Patients started on calcium channel blockers (CCBs) were not excluded in previous reports, so the 2016-17 results have been recalculated. **2.** Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend.



NS 11: Patient quality of life should be recorded: Overview

Audit standard

Ninety per cent of adult patients who have PAH or CTEPH and who have at least one consultation (inpatient, day case, or outpatient) in the last year should have at least one quality of life questionnaire recorded in the last year.



National: 96%

Target: 90%

Met at: 7 of 7 centres

Trend: All centres met for first time

“A PHA UK survey identified that quality of life is of importance to patients and therefore measuring and completing this indicator at each visit is an integral part of our assessment.”

Alan Greenhalgh
Research Nurse

Why is this important?

Assessment of the quality of life of patients with pulmonary hypertension is best achieved by asking patients to complete a questionnaire. Changes in quality of life can then be monitored in response to treatment.

PHA UK have championed the introduction of this standard.



NS 11: Patient quality of life should be recorded: Results

Table NS 11: Proportion of cases meeting National Standard 11, Great Britain, 2016-19

Adults' PH centre (Target = 90%)	Audit year			Trend ² 17-18 to 18-19	
	2016-17	2017-18	2018-19	<i>p</i> ≤0.05	
	Per cent	Per cent	Per cent		
Golden Jubilee	94	91	94	Static	—
Imperial College	79	99	95	Down	▼
Newcastle	85	98	100	Up	▲
Royal Brompton and Harefield	95	97	93	Down	▼
Royal Free	92	97	98	Static	—
Royal Papworth	93	88	93	Up	▲
Sheffield	88	97	99	Up	▲
National total	89	96	96	Static	—

[Methodology](#)



Key finding

All PH centres met the standard for the first time

Key:

▲ = Upwards trend
— = Static trend
▼ = Downward trend

Target met	n
Target not met	n
<i>p</i> ≤0.05	See Glossary

Notes: 1. Definition: 90 per cent of adult patients who have PAH or CTEPH and who have at least one consultation (inpatient, day case, or outpatient) in the last year should have at least one quality of life questionnaire recorded in the last year. 2. Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend.



NS 12: Patients receiving a PH drug should have an annual consultation: Overview

Audit standard

Ninety per cent of patients receiving a pulmonary hypertension drug therapy should have had a consultation (inpatient, day case or outpatient) within the last 13 months¹.

Why is this important? Specialist PH centres provide follow-up for patients in whom they are prescribing specialist drug therapies. It is crucial that the effects of these drug therapies adequacy of response to treatment and side effects. This monitoring should be carried out at least annually by prescribers. Note that 13 months is chosen to represent one year so as to allow some variation in the timing of patients who only have an annual appointment with their PH centre.



National: 96%

Target: 90%

Met at: 8 of 8 centres

Trend: All met for last three years

New changes to National Standard 12

To take account of more virtual consultations for sick patients unable to travel to their designated PH centre, the target for National Standard 12 has been reduced from 95 to 90 per cent.

Notes: 1. This standard is based on the NHS England PH service specification [A11/S/a](#) 2013-14: "Patients treated with disease targeted therapy will have lifelong follow up within the PH service. The PH centre will identify those patients suitable for shared care and ensure effective communication with shared care PH centres to plan patient reviews. All such patients will be reviewed at least once each year by the visiting PH specialist or at the PH centre." (p.6).



NS 12: Patients receiving a PH drug should have an annual consultation: Results

Table NS 12: Proportion of cases meeting National Standard 12, Great Britain, 2016-19

PH centre (Target = 90%)	Audit year			Trend ² 17-18 to 18-19	
	2016-17	2017-18	2018-19		
	Per cent	Per cent	Per cent	$p \leq 0.05$	
Golden Jubilee	95	93	97	Up	▲
Great Ormond Street	96	95	95	Static	—
Imperial College	94	95	95	Static	—
Newcastle	97	98	99	Static	—
Royal Brompton and Harefield	96	96	97	Static	—
Royal Free	95	98	97	Static	—
Royal Papworth	97	96	96	Static	—
Sheffield	97	95	94	Static	—
National total	96	96	96	Static	—

[Changes](#)

[Methodology](#)



Key finding

All PH centres met the standard for the third consecutive year

Key:

- ▲ = Upwards trend
- = Static trend
- ▼ = Downward trend

Target met n

Target not met n

$p \leq 0.05$ See [Glossary](#)

Notes: 1. Definition: 90 per cent of patients receiving a pulmonary hypertension drug therapy should have had a consultation (inpatient, day case or outpatient) within the last 13 months. 2. Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend.



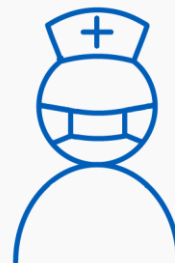
NS 13: Waiting times for PEA should be <4 months: Overview

Audit standard

90 per cent of adult patients who undergo a pulmonary endarterectomy (PEA) should have waited less than 4 months from diagnosis of chronic thromboembolic pulmonary hypertension by the multidisciplinary team (MDT) meeting at the specialist PH centre¹.

“Patients are being discussed sooner at the PEA MDT but the number of operations did not increase due to the hospital relocation in 2019. In this new **Royal Papworth** hospital we will be able to perform more operations.”

Joanna Pepke-Zaba
Consultant Physician



National: 11%

Target: 90%

Met at: 0 of 7 centres

Trend: Never met

Why is this important?

PEA is an operation which is performed for the treatment of selected patients with CTEPH. This may result in marked clinical improvement for patients.

This measurement looks at the time it takes from diagnosis at a specialist PH centre to surgery at Royal Papworth, the single PEA centre in the UK.



NS 13: Waiting times for PEA should be <4 months: Results

Table NS 13: Proportion of cases meeting National Standard 13, Great Britain, 2016-19

Adults' PH centre (Target = 90%)	Audit year			Trend ² 17-18 to 18-19	
	2016-17	2017-18	2018-19		
	Per cent	Per cent	Per cent	$p \leq 0.05$	
Golden Jubilee	0	9	23	Static	—
Imperial College	14	6	10	Static	—
Newcastle	33*	8	0*	Static	—
Royal Brompton and Harefield	0*	10	0*	Static	—
Royal Free	13	17	0	Static	—
Royal Papworth	27	36	23	Static	—
Sheffield	2	7	6	Static	—
National total	13	18	11	Static	—

[Methodology](#)

[Information](#)



Key finding

Only 11 per cent of patients had surgery within 4 months

Key:

- ▲ = Upwards trend
- = Static trend
- ▼ = Downward trend

Target met n

Target not met n

$p \leq 0.05$ See [Glossary](#)

Notes: * = Fewer than 10 cases in the denominator. **1.** Definition: 90 per cent of adult patients who undergo a pulmonary endarterectomy should have waited less than 4 months from diagnosis of chronic thromboembolic pulmonary hypertension by the multidisciplinary team meeting at the pulmonary hypertension specialist PH centre. **2.** Variation between audit years can occur in a 'static' trend. Where fewer cases are involved, a greater degree of variation is possible within a 'static' trend.



National Audit of Pulmonary Hypertension

Survival analysis:

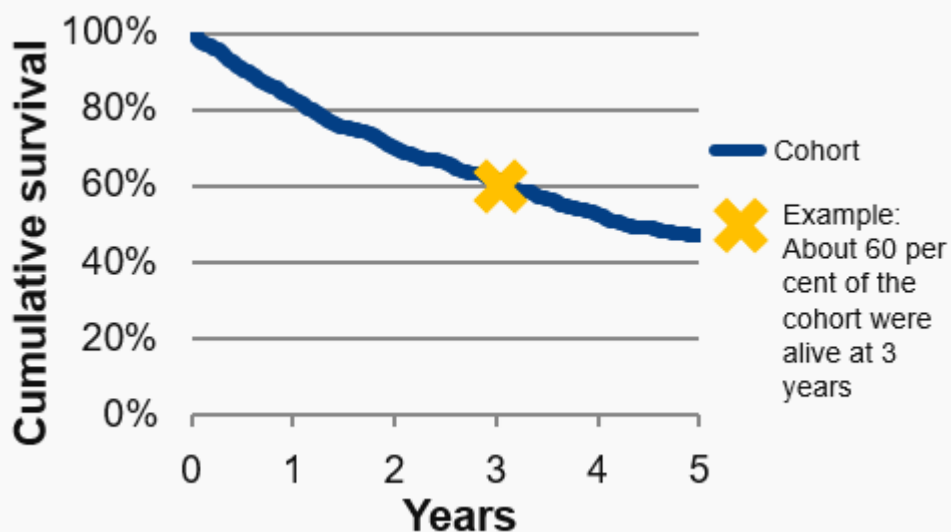
Deaths in the longitudinal cohort



Survival analysis: Overview

Why is this important? PH may be a life shortening condition. One of the aims of treatment is to prolong life as well as to improve its quality. Measurement of survival is used as a clinical outcome; it can be used to assess the effect of treatment on many life threatening diseases.

How is survival reported? Kaplan Meier curves are used as a way of graphically displaying an estimate of what proportion of people will still be in a group a number of days after a chosen start date. For example, it provides an estimate of the percentage of patients that will still be alive x years after they received a diagnosis (see below).



How is survival measured?

Survival is measured from the *earliest* diagnosis date for the patient's *latest* diagnosis. 98 per cent of patients keep the same PH diagnosis.

Who is included?

Only adult patients¹ within the cohort for longitudinal analysis are included (those whose first referral was on or after 1 April 2009²). Patients whose diagnosis date is more than one year prior to first referral are excluded.

The survival curves are stopped if the number of patients used to calculate the cumulative survival drops below 50. This is to prevent the curve from artificially flattening due to patients whose mortality status cannot be confirmed.

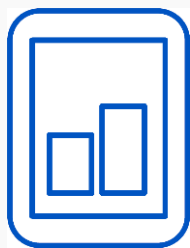
How is mortality data collected?

Date of death is submitted directly to NAPH by the PH centres, then supplemented by mortality data from the Office for National Statistics (ONS).

Patients submitted by the Scottish PH centre (Golden Jubilee) cannot be traced at ONS or the National Records of Scotland because they are pseudo-nymised in NAPH². Scotland are not included in the lung disease or left heart disease survival curves due to difficulties in obtaining death data for those patients.



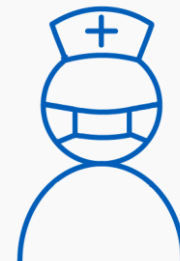
Survival analysis: Summary



[Chart S 1](#) and [Table S 1](#) show that survival for patients with **congenital heart disease** APAH is better than for all other PAH types. Similarly, [Chart S 2](#) shows that survival for **IPAH** patients without comorbidities is better than for those with comorbidities.

Although Chart S 2 suggests that **connective tissue disease** APAH survival is worse for those with scleroderma, [Chart S 3](#) shows that the results are not consistently different at the 95% significance level. This may be due to the relatively small size of the patient groups.

[Chart S 4](#) shows that patients with **CTEPH** have the highest chance of survival of all types of PH, particularly those who have undergone an operation for the condition. Note that patient who have undergone surgery are younger than those who are not operated. Chart S 4 also shows that **lung disease** has the lowest survival rate with only half likely to survive after two years following diagnosis.



[Chart S 5](#) shows that survival for CTEPH patients that are **operated** is better than for those that are not operated.

Breakdowns of the characteristics of each diagnostic group are provided in Table A 1 to 8 in the [Appendix](#) covering:

- Patient demographics (age, sex)
- WHO functional class and six-minute walk test
- Quality of life (emPHasis-10 score)
- Haemodynamics (blood pressure)
- Lung function test results
- Therapy type (mono, combination or none)



A comparison of IPAH survival before and after the **2019 IPAH Review** are provided from Table A 9 onwards in the [Appendix](#).

Survival for IPAH patients is split into the following categories in the Supplementary Survival Report:

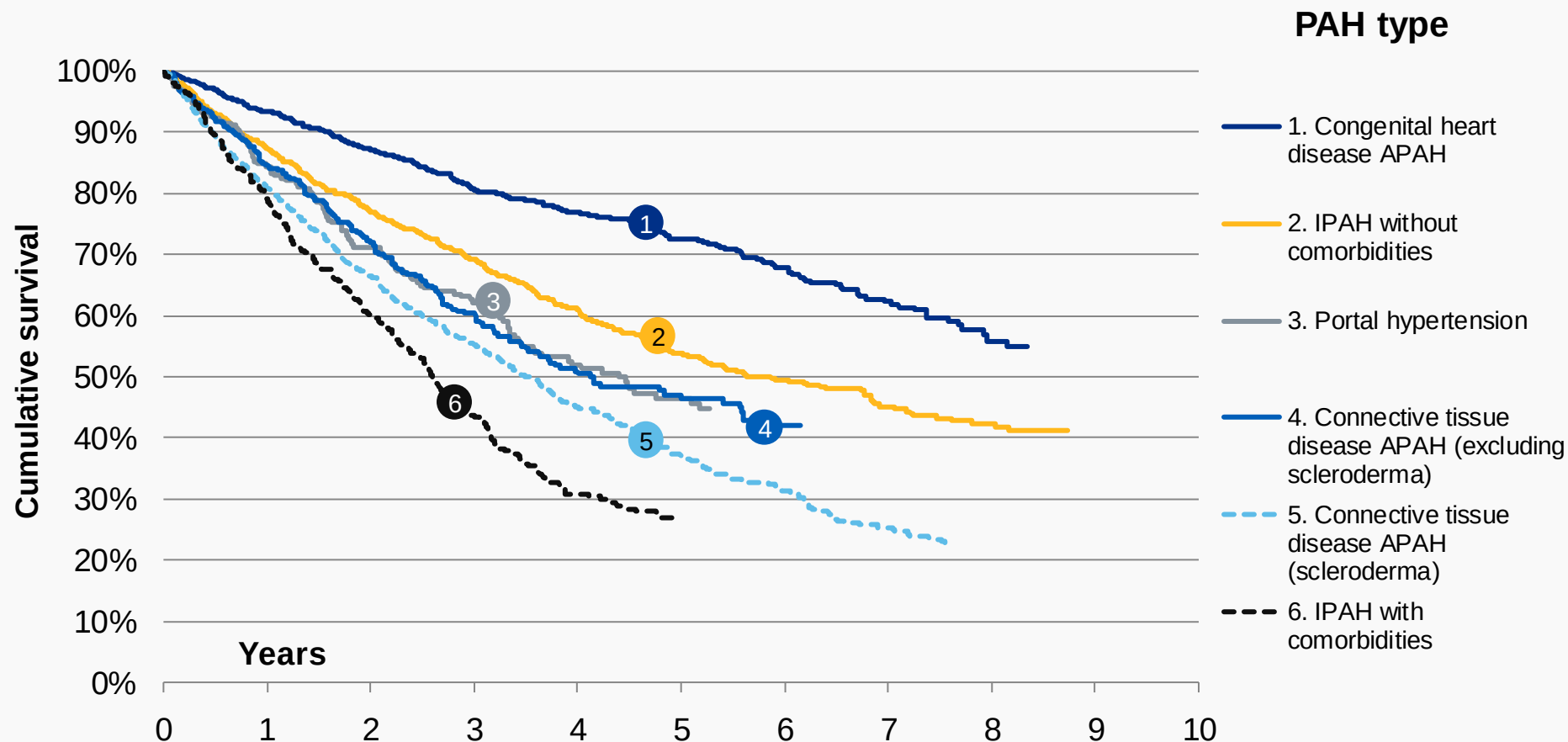
- Sex
- Age quintile at diagnosis
- WHO functional class



Survival analysis: PAH type

Survival curves

Chart S 1: Survival curves from diagnosis by PAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19



Notes: 1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.

2. Patients with connective tissue disease where scleroderma is not specified are not included in the chart.

3. See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Survival analysis: PAH type

Annual survival

Table S 1: Post-diagnosis survival by PAH type,

Adult patients in the longitudinal cohort, Great Britain, 2009-19

Years Elapsed	IPAH				Connective tissue disease APAH				Congenital heart disease APAH		Portal hypertension	
	Without comorbidities		With comorbidities		Scleroderma		Excluding scleroderma		Patients	% survival	Patients	% survival
	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival				
0	1,132	100	385	100	1,045	100	381	100	1,117	100	278	100
1	880	87	285	79	745	81	287	85	941	94	211	84
2	675	77	189	60	535	66	212	72	776	87	161	71
3	537	69	116	43	385	55	151	60	610	81	128	62
4	407	61	67	31	256	45	105	50	491	77	83	52
5	288	54	–	–	168	37	71	47	376	73	57	46
6	206	49	–	–	110	31	52	42	223	68	–	–
7	135	45	–	–	67	25	–	–	142	62	–	–
8	78	42	–	–	–	–	–	–	58	56	–	–
9	–	–	–	–	–	–	–	–	–	–	–	–

Key finding

Survival for patients with congenital heart disease APAH is better than for all other PAH types (diverging from all types from 105 days onwards at 95% confidence).

Notes: **1. IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH. **2.** Patients with connective tissue disease where scleroderma is not specified are not included in the table. **3.** See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.

Breakdowns of the characteristics of each diagnostic group are available in the [Appendices](#):

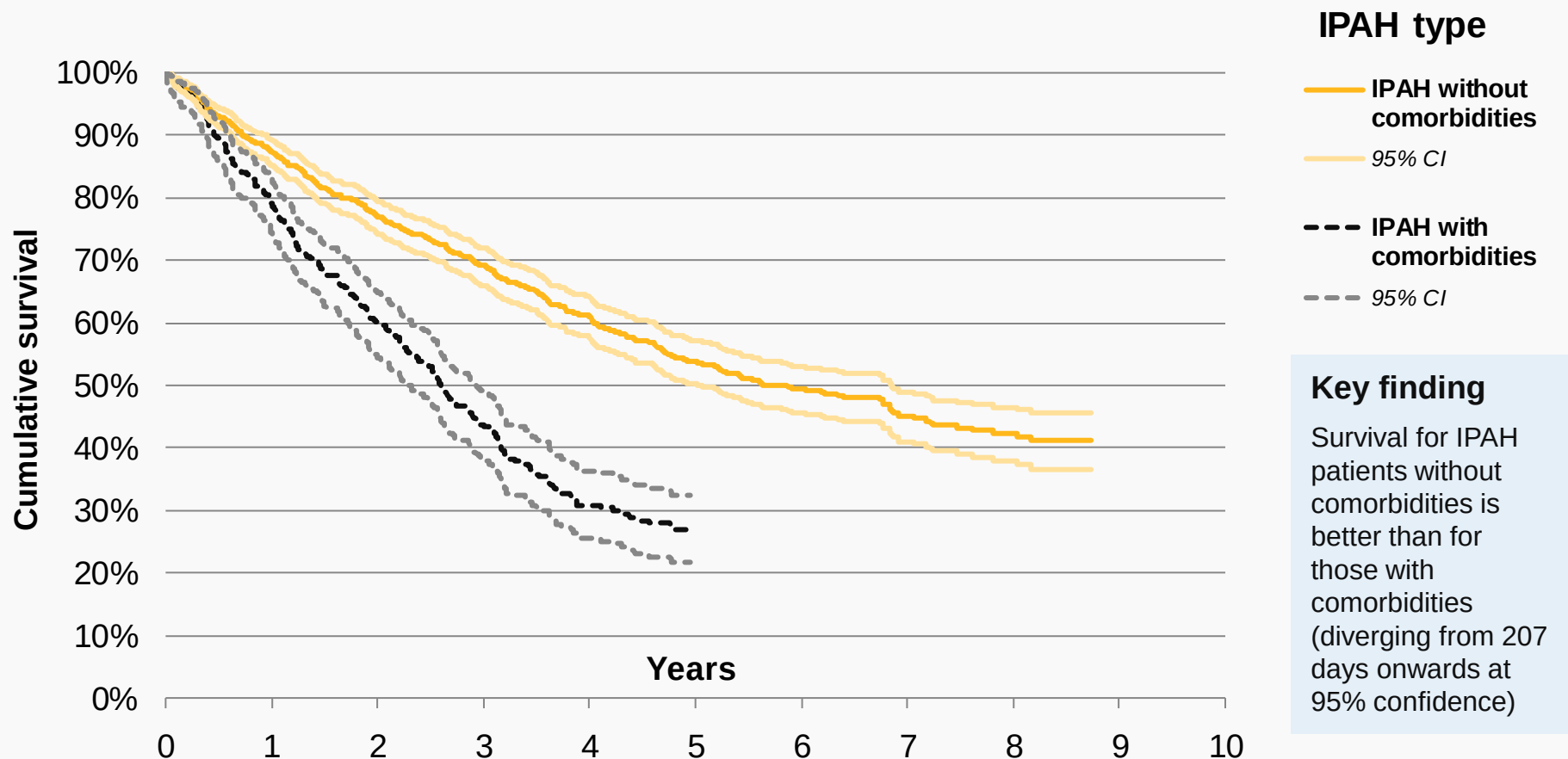
- [Patient demographics](#) (age, sex)
- [PH characteristics](#) including WHO functional class, six-minute walk test, quality of life (emPHasis-10 score) and haemodynamics (blood pressure)
- [Lung function test](#) results
- [Therapy type](#) (mono, combination or none)



Survival analysis: IPAH type

Survival comparison

Chart S 2: Survival curves from diagnosis by IPAH type: comparison,
Adult patients in the longitudinal cohort, Great Britain, 2009-19



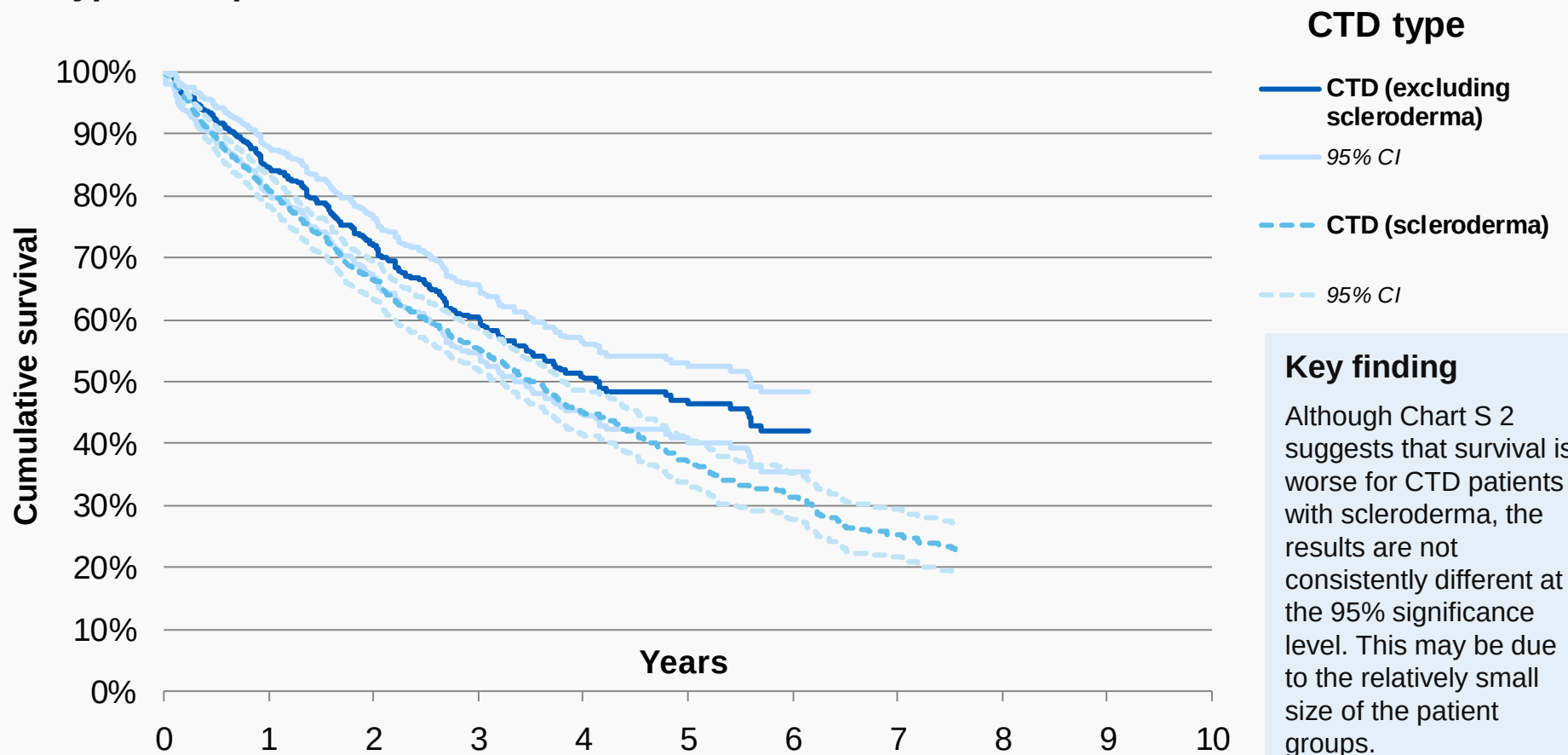
Notes: 1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Survival analysis: CTD type

Survival comparison

Chart S 3: Survival curves from diagnosis by connective tissue disease APAH type: comparison, Adult patients in the longitudinal cohort, Great Britain, 2009-19



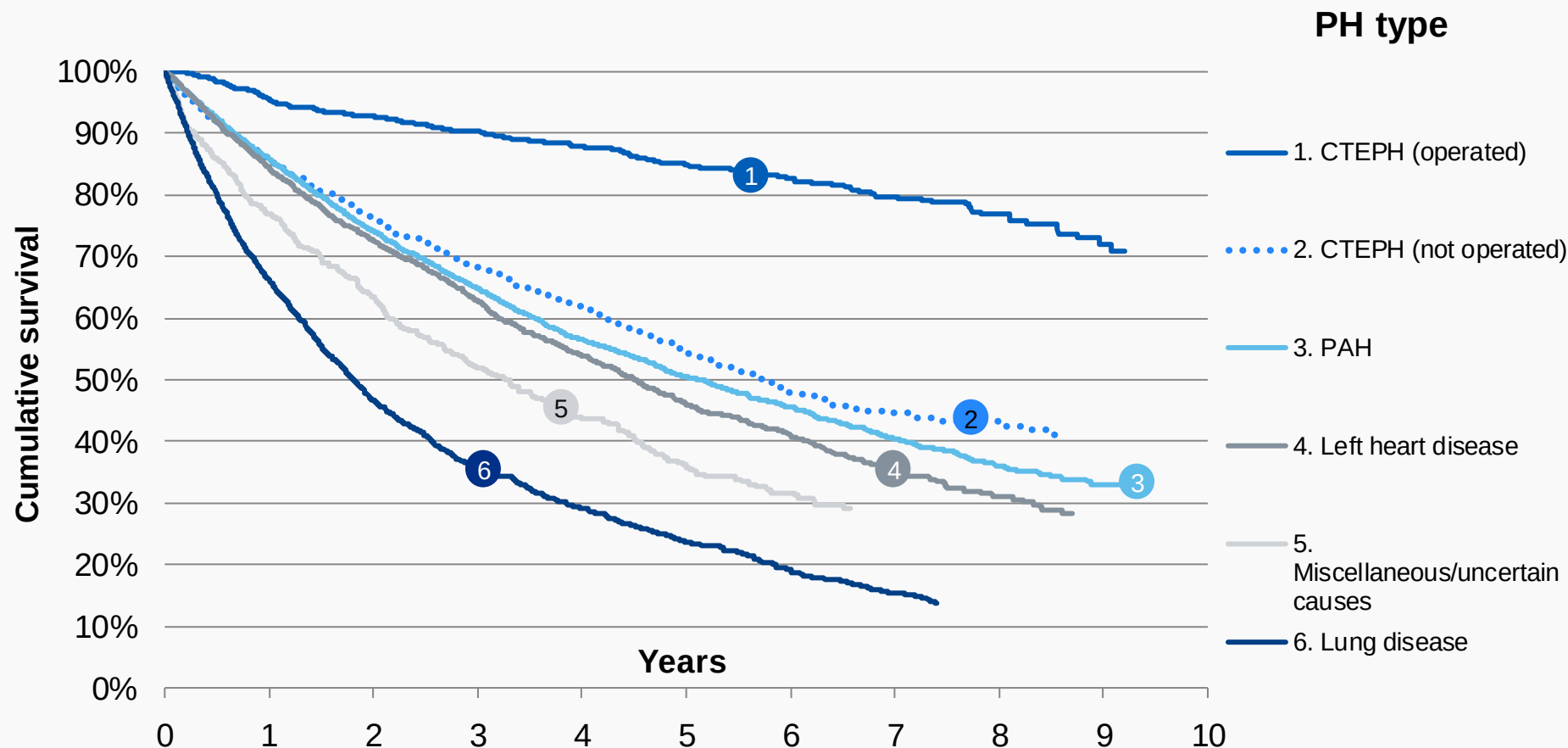
Notes: 1. See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.
2. Patients with connective tissue disease where scleroderma is not specified are not included in the chart.



Survival analysis: PH type

Survival curves

Chart S 4: Survival curves from diagnosis by PH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19



Notes: 1. The **PAH** group includes patients with congenital heart disease APAH, connective tissue disease APAH, IPAH, portal hypertension APAH and other PAH sub-groups.
2. See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Survival analysis: PH type

Annual survival

Table S 2: Post-diagnosis survival by PH type,

Adult patients in the longitudinal cohort, Great Britain, 2009-19

Years Elapsed	PAH		Left heart disease		Lung disease		CTEPH				Miscellaneous/ uncertain causes	
							Operated		Not operated			
	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival
0	4,812	100	3,265	100	2,093	100	1,237	100	1,583	100	528	100
1	3,688	86	2,378	84	1,178	66	1,116	95	1,094	86	366	77
2	2,792	74	1,756	73	717	47	940	93	822	76	254	64
3	2,112	65	1,213	63	468	36	792	90	626	68	184	52
4	1,546	57	827	54	311	29	638	88	462	62	135	44
5	1,098	51	545	46	185	23	498	85	322	54	96	36
6	712	46	366	41	114	19	392	83	208	48	69	32
7	442	40	220	36	62	15	284	80	129	45	–	–
8	220	36	93	31	–	–	159	77	83	42	–	–
9	85	33	–	–	–	–	71	72	–	–	–	–

Key findings

Patients with CTEPH have the highest chance of survival of all types of PH, particularly those who have undergone an operation for the condition. Note that patient who have undergone surgery are younger than those who are not operated.

Lung disease has the lowest survival rate with only half likely to survive after two years following diagnosis.

Breakdowns of the characteristics of each diagnostic group are available in the [Appendices](#):

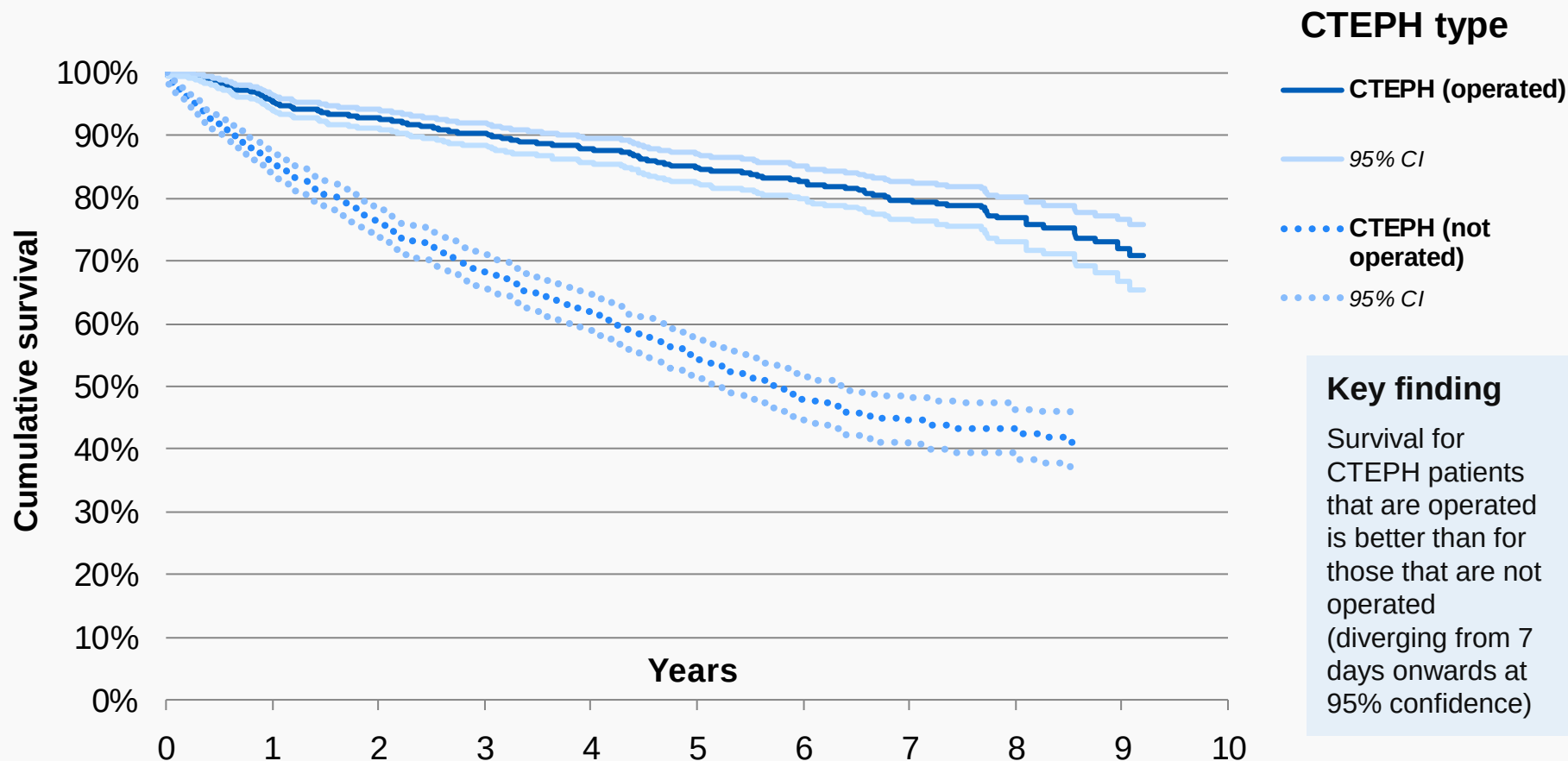
- [Patient demographics](#) (age, sex)
- [PH characteristics](#) including WHO functional class, six-minute walk test, quality of life (emPHasis-10 score) and haemodynamics (blood pressure)
- [Lung function test](#) results
- [Therapy type](#) (mono, combination or none)



Survival analysis: CTEPH type

Survival comparison

Chart S 5: Survival curves from diagnosis by CTEPH type: comparison,
Adult patients in the longitudinal cohort, Great Britain, 2009-19



National Audit of Pulmonary Hypertension

Reference tables:

Patients and drug therapies



Reference tables: Summary



[Chart R 1](#) shows that the number of active patients and active drug therapies has consistently risen since 2004. A similar increase in managed patients with and without PAH/CTEPH is shown in [Table R 2](#).

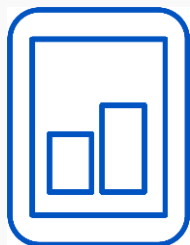
[Table R 1](#) shows that this growth is evident across most PH centres since 2010, though the trend is less pronounced at Great Ormond Street and Royal Papworth.

Variations in diagnosis types between PH centres are shown in [Table R 3](#) and [Table R 4](#).

[Table R 5](#) shows that Sildenafil is prescribed most frequently. Prescriptions of all PH drugs are increasingly, except for Bosentan and Treprostinil. The most marked increases in prescription numbers are for Sildenafil, Tadalafil, Ambrisentan and Macitentan. Similar patterns are evident for CTEPH patients ([Table R 6](#)) and IPAH patients without comorbidities ([Table R 7](#)).



[Table R 8](#) shows that the number of patients diagnosed and living with CTEPH is increasing, as is the proportion of CTEPH patients having active therapy since 2011. The proportion of CTEPH patients discharged for reasons other than death is around 16 per cent.



[Chart R 2](#) suggests a shortening of the length between initial diagnosis of CTEPH and PEA performed between 2012-13 and 2019-19.

[Table R 9](#) shows that there has been an increase in active drug therapies that are combination therapies since 2010, though this trend is less pronounced for CTEPH patients.

[Table R 10](#) and the accompanying footnote show that there were only a small number of transplants (lung or heart/lung) during 2018-19.

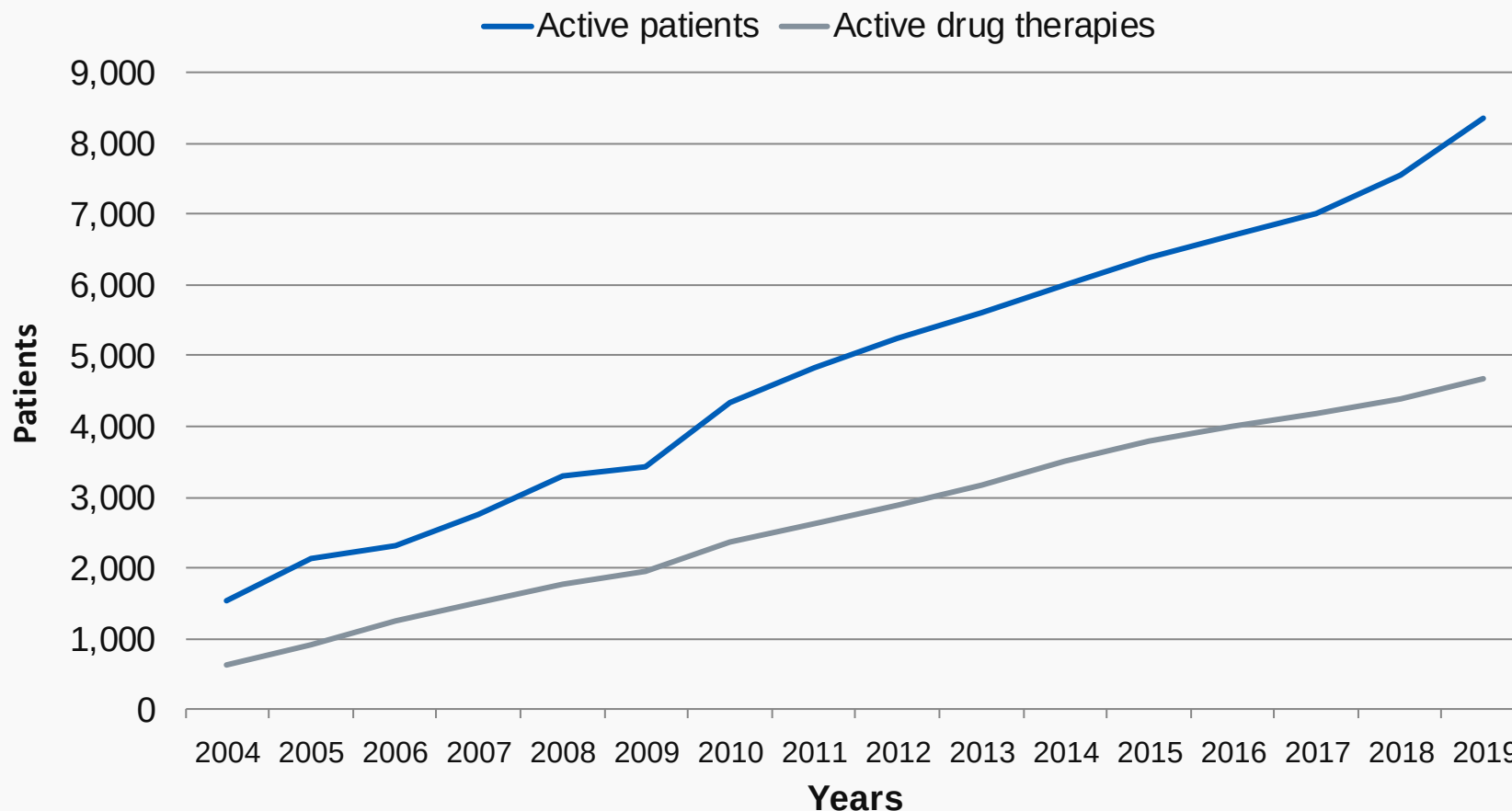


Reference tables:

Active patients and therapies

[Glossary](#)

Chart R 1: Number of active patients and active drug therapies on 31 March, Great Britain, 2004-19



Notes: 1. **Active patients** have a referral active on the 31 March of the stated year (see [Glossary: Patient cohorts](#)). 2. **Active drug therapies** represent patients with at least one pulmonary hypertension drug therapy being prescribed on the 31 March of the appropriate year (see [Glossary: Patient cohorts](#)). 3. The data for 2004-2009 was collected before the Audit began. It was collected by the National Pulmonary Hypertension Centres of the UK and Ireland Physicians Committee.



Reference tables: Active patients

[Glossary](#)

Table R 1: Number of active patients on 31 March by PH centre, Great Britain, 2010-19

PH centre	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Total	4,341	4,835	5,254	5,601	6,007	6,382	6,695	6,998	7,550	8,351
Golden Jubilee	350	369	401	441	498	534	578	600	649	669
Great Ormond Street	278	266	304	301	301	294	298	299	307	311
Imperial College	650	713	793	842	874	933	919	897	946	1,034
Newcastle	280	290	300	313	316	330	333	326	339	361
Royal Brompton & Harefield	502	602	634	693	691	691	651	749	764	829
Royal Free	629	757	860	905	905	1,000	1,084	1,154	1,397	1,774
Royal Papworth	705	751	815	900	964	1,061	1,161	1,229	1,212	1,190
Sheffield	947	1,087	1,147	1,206	1,458	1,539	1,671	1,744	1,936	2,183

Key findings

The number of active patients has consistently increased across most PH centres from 2010 to 2019.



Reference tables: Managed patients

Table R 2: Number of managed patients during the audit year, Great Britain, 2009-19

Audit year	Managed patients	Managed patients with PAH or CTEPH	
		n	%
2009-10	5,705	3,280	57
2010-11	6,411	3,762	59
2011-12	7,004	4,124	59
2012-13	7,503	4,464	59
2013-14	8,121	4,876	60
2014-15	8,636	5,245	61
2015-16	9,024	5,509	61
2016-17	9,617	5,803	60
2017-18	9,922	5,979	60
2018-19	10,572	6,069	57

Key findings

The number of managed patients with PH continues to increase.

[Glossary](#)



Reference tables:

Diagnosis type of active patients

[Glossary](#)

Table R 3: Percentage of active patients on 31 March 2019 by PH centre and diagnosis type, Great Britain, 2019

Diagnosis type	PH centre								
	National total	Golden Jubilee	Great Ormond Street	Imperial College	Newcastle	Royal Brompton and Harefield	Royal Free	Royal Papworth	Sheffield
	%	%	%	%	%	%	%	%	%
Total	8,351	669	311	1,034	361	829	1,774	1,190	2,183
PAH	43	42	43	43	57	59	26	25	57
Pulmonary veno-occlusive disease and/or pulmonary capillary haemangiomatosis	1	2	1	0	0	1	0	1	0
Left heart disease	5	1	4	5	0	1	12	4	2
Lung disease	4	12	25	2	1	3	4	1	4
CTEPH	23	22	0	25	29	16	15	47	23
...operated	12	10	0	14	13	5	7	27	10
...not operated	12	12	0	11	16	11	8	19	13
Miscellaneous/uncertain causes	2	3	5	1	1	8	1	2	2
Not pulmonary hypertension	16	14	14	20	3	2	35	15	6
No final diagnosis possible	1	0	0	0	1	8	2	0	0
No diagnosis	5	5	9	4	8	2	5	6	6



Reference tables:

PAH type of active patients

[Glossary](#)

Table R 4: Percentage of active patients on 31 March 2019 by PH centre and PAH type, Great Britain, 2019

PAH type	PH centre								
	National total	Golden Jubilee	Great Ormond Street	Imperial College	Newcastle	Royal Brompton and Harefield	Royal Free	Royal Papworth	Sheffield
	%	%	%	%	%	%	%	%	%
Total	3,551	279	134	442	207	491	456	296	1,246
IPAH	30	39	25	45	44	16	16	39	29
...without comorbidities	26	37	17	44	42	15	13	39	21
...with comorbidities	4	2	7	1	2	1	3	0	8
Connective Tissue Disease APAH	22	23	2	11	18	15	49	6	25
...scleroderma	15	12	1	5	12	9	38	6	18
...excluding scleroderma	7	11	1	6	6	7	11	0	7
Portal Hypertension APAH	4	10	2	4	5	2	6	3	3
Congenital Heart Disease APAH	36	26	68	31	27	64	13	23	40
PAH - Other	5	1	2	5	3	2	12	16	2
PAH - No sub diagnosis	2	0	1	4	2	0	4	13	0



Reference tables:

Drug prescriptions

Table R 5: Number of drug prescriptions on 31 March, Great Britain, 2010-19

Drug	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Sildenafil	1,518	1,754	2,100	2,386	2,670	2,980	3,131	3,264	3,284	3,356
Tadalafil	3	4	31	107	144	190	257	338	498	709
Bosentan	1,031	1,113	1,141	1,134	1,178	1,002	833	710	624	597
Ambrisentan	125	274	324	430	514	540	594	616	652	707
Sitaxsentan	100	2	-	-	-	-	-	-	-	-
Macitentan	-	-	-	-	-	296	535	720	979	1,139
Iloprost	109	107	111	123	151	154	149	152	159	193
Treprostinil	88	69	60	55	44	40	32	27	26	27
Epoprostenol	46	57	67	84	87	99	115	120	138	150
CCB* for vaso-reactive PAH	37	34	41	50	56	56	59	64	74	79
Riociguat	-	-	-	-	-	14	73	110	148	180
Selexipag	-	-	-	-	-	-	-	0	3	41
Unknown – Drug trial	106	125	123	121	245	239	244	237	222	213
Unknown – Not known	12	10	8	9	8	8	8	8	8	8

Notes: * Calcium channel blocker.

1. Drug prescriptions include monotherapy and combination therapy.

2. Sitaxsentan was withdrawn from use in late 2010.



Reference tables:

Drug prescriptions for CTEPH patients

Table R 6: Number of drug prescriptions for CTEPH patients on 31 March,
Great Britain, 2010-19

Drug	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Sildenafil	317	381	434	490	532	611	639	652	674	714
Tadalafil	0	0	1	11	18	21	26	34	55	71
Bosentan	180	174	162	153	146	126	90	70	45	46
Ambrisentan	10	20	32	53	66	75	72	65	60	53
Macitentan	-	-	-	-	-	32	55	74	138	178
Iloprost	11	14	10	12	12	15	9	7	7	8
Treprostinil	4	4	3	2	2	0	0	0	0	0
Epoprostenol	1	2	3	4	3	3	4	2	1	2
Riociguat	-	-	-	-	-	8	65	99	117	138
Selexipag	-	-	-	-	-	-	-	0	0	3
Unknown – Drug trial	2	2	3	4	19	14	7	7	7	6
Unknown – Not known	0	1	2	1	1	1	1	1	1	1

Notes:

1. Drug prescriptions include monotherapy and combination therapy
2. For this report the inclusion criteria has been extended to include all patients. In previous reports only patients in the longitudinal cohort were included (see [Glossary: Patient cohorts](#)).



Reference tables:

Drug prescriptions for IPAH patients

Table R 7: Number of drug prescriptions for IPAH patients without comorbidities on 31 March, Great Britain, 2010-19

Drug	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Sildenafil	337	406	478	526	541	583	588	578	577	558
Tadalafil	2	2	9	25	37	49	64	85	131	201
Bosentan	214	235	259	257	245	207	161	130	118	105
Ambrisentan	38	72	94	128	166	174	194	214	215	252
Sitaxsentan	24	0	-	-	-	-	-	-	-	-
Macitentan	-	-	-	-	-	102	178	216	263	292
Iloprost	40	41	43	58	66	65	65	69	75	85
Treprostinil	46	37	36	34	26	26	21	19	18	19
Epoprostenol	27	31	34	44	51	56	62	68	81	87
CCB* for vaso-reactive PAH	30	26	31	38	40	43	46	51	61	65
Riociguat	-	-	-	-	-	6	8	8	23	22
Selexipag	-	-	-	-	-	-	-	0	1	21
Unknown – Drug trial	34	44	50	54	162	162	169	166	157	153
Unknown – Not known	4	4	0	0	0	0	0	0	0	0



Reference tables: CTEPH

[Glossary](#)

Table R 8: Patients with CTEPH and new CTEPH diagnoses, Great Britain, 2009-19

Patients with CTEPH...	Audit year (patients)									
	2009 -10	2010 -11	2011 -12	2012 -13	2013 -14	2014 -15	2015 -16	2016 -17	2017 -18	2018 -19
• alive on 31 March at the end of audit year	805	996	1,185	1,370	1,530	1,725	1,914	2,054	2,263	2,492
• with an active referral ¹ at the end of audit year	754	907	1,051	1,182	1,305	1,442	1,578	1,638	1,784	1,962
• newly-diagnosed with CTEPH during audit year	237	264	269	279	279	323	343	319	375	387
Patients with CTEPH...	End of audit year (per cent)									
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
• in the cohort for longitudinal analysis and having active drug therapy on 31 March ²	48	45	42	41	42	43	44	45	45	47
Patients with CTEPH...	Diagnosed before end of audit period (per cent)									
	2009 -10	2009 -11	2009 -12	2009 -13	2009 -14	2009 -15	2009 -16	2009 -17	2009 -18	2009 -19
• in the cohort for longitudinal analysis ³ and discharged from PH service within 5 years (excluding where reason for PH discharge is death) ⁴	-	-	-	-	-	17	17	18	17	16

Notes:

1. **Active patients** have a referral active on the 31 March of the stated year (see [Glossary: Patient cohorts](#)).

2. For the 10th Annual Report filter added to restrict to active patients only.

3. The **cohort for longitudinal analysis** includes those whose first referral was on or after 1 April 2009 (see [Glossary: Patient cohorts](#)).

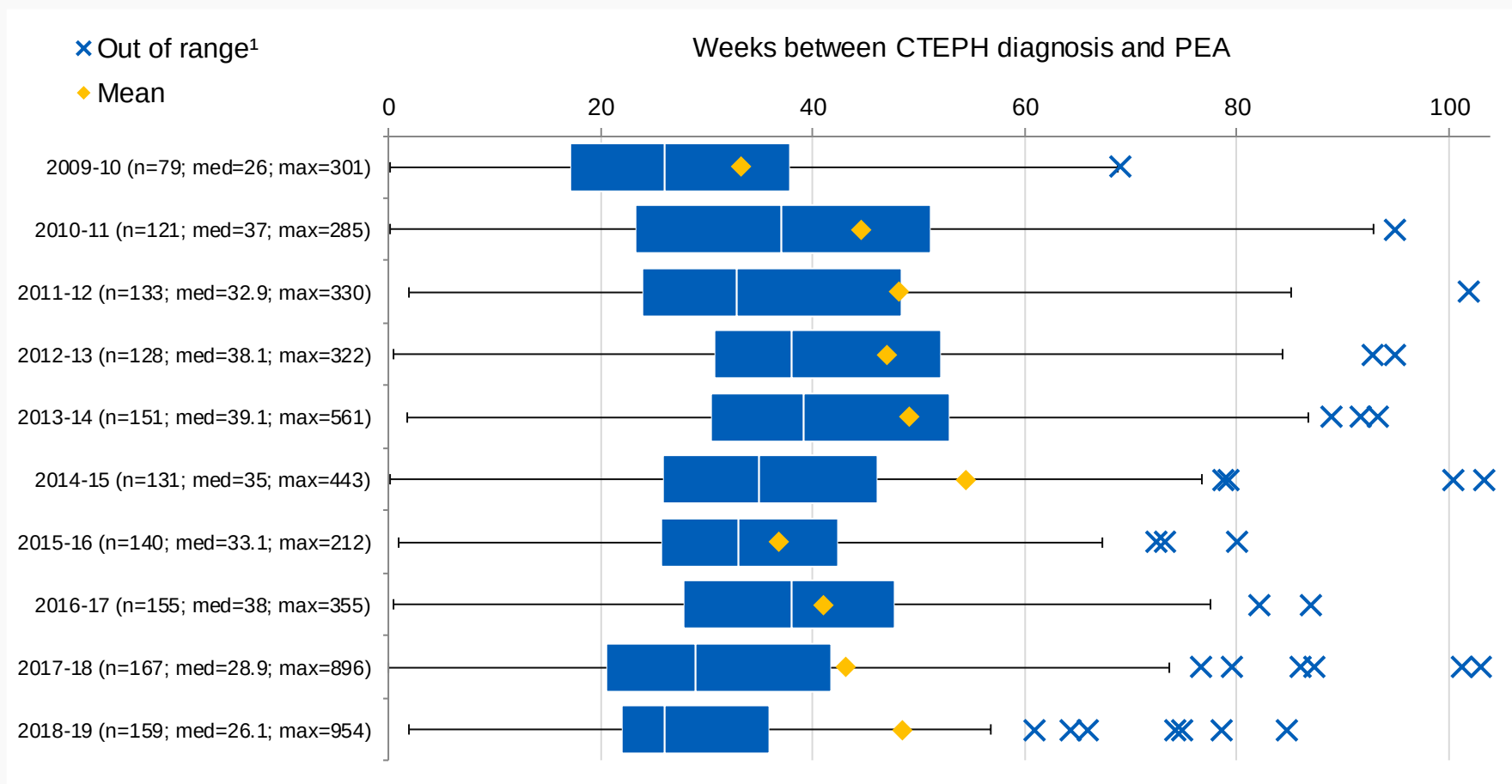
4. For the 10th Annual Report filter added to stop counting patients whose reason for discharge is death from the numerator.



Reference tables: Interval to PEA

[Glossary](#)

Chart R 2: Number of weeks between initial diagnosis of CTEPH and PEA performed, Great Britain, 2009-19



Notes: 1. Intervals outside the interquartile range (IQR) by more than 1.5 times the IQR are shown as 'out of range' (blue cross). Not all out of range values are displayed on the plot. The maximum interval for each audit year is shown on the y axis (left).



Reference tables:

Transplants and therapy types

[Glossary](#)

Table R 9: Percentage of active drug therapies¹ that are mono-, dual or triple therapies² on 31 March, Great Britain, 2010-19

Year	All patients			CTEPH			IPAH without comorbidities		
	Mono	Dual	Triple+	Mono	Dual	Triple+	Mono	Dual	Triple+
2010	72	25	3	81	18	1	56	37	7
2011	71	26	2	82	17	1	53	41	6
2012	68	30	3	82	17	1	45	48	7
2013	64	32	3	79	20	1	42	50	8
2014	64	33	3	78	21	1	39	52	9
2015	61	36	3	78	21	1	34	57	9
2016	59	37	4	80	19	1	32	58	11
2017	58	38	4	81	19	0	29	58	13
2018	55	41	4	79	21	0	28	59	14
2019	53	42	5	77	22	0	25	57	18

Notes:

1. Active drug therapies represent patients with at least one pulmonary hypertension drug therapy being prescribed on the 31 March of the appropriate year (see [Glossary: Patient cohorts](#)).

2. The methodology to produce therapy type has been updated for the 10th Annual Report. Therapy type is now derived from the number of concurrent **drug classes** prescribed on the latest recorded therapy date, rather than the number of drugs. See the [Glossary: PH Characteristics \(1\)](#) slide for details of how drug class is defined.

Table R 10: Number of transplants³ (lung or heart/lung) in 2018-19, Great Britain, 2018-19

Description	Number of transplants
Transplants	6



National Audit of Pulmonary Hypertension

Appendices:
Tables and charts



Appendices: Navigation



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The following documents are published alongside the main report

- Supplementary Survival Analysis: Sex, Age and WHO Functional Class
- Tables and Charts (Excel)
- Open Data (csv)
- Data Quality Report
- Dana Point Classification
- Staff Involved in Audit



Appendices: PAH type Demographics

[Glossary](#)

Table A 1: Demographics by PAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Diagnostic group	Patients	Age at diagnosis				Female	Days after diagnosis on which the cohort had 50 per cent mortality
	Number	Lower quartile	Median	Upper quartile	Mean	%	
IPAH							
...without comorbidities	1,132	45	60	72	58	65	5 years and 293 days
...with comorbidities	385	63	70	77	68	53	2 years and 218 days
Connective tissue disease APAH							
...scleroderma	1,045	59	67	73	66	83	3 years and 193 days
...excluding scleroderma	381	49	61	71	59	76	4 years and 43 days
Congenital heart disease APAH	1,117	33	45	60	47	65	<i>Not reached at 8 years and 124 days</i>
Portal hypertension	278	46	54	62	53	44	4 years and 145 days

Notes:

- IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
- Patients with connective tissue disease where scleroderma is not specified are not included in the table.
- See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Appendices: PAH type PH characteristics

[Glossary](#)

Table A 2: PH characteristics at diagnosis by PAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Parameters	I = least severe IV = most severe				Further = healthier		0=best QoL 50=worst		Norm~14. 25+ =PH		Norm ~8		Norm= 0 to 7		Norm= ≤3		Norm= 65 to 75		Norm= 2.5 to 4.2	
Diagnostic group	WHO functional class				Six-minute walk test (metres)		Quality of life: emPHasis-10 score (0-50)		Mean pulmonary artery pressure (PAP) (mm Hg)		Mean pulmonary wedge pressure (PWP) (mm Hg)		Mean right atrial pressure (RAP) (mm Hg)		Mean pulmonary vascular resistance (PVR) (Wood units)		Mixed venous oxygen saturation (SvO2) (%)		Cardiac index (L/min/m ²)	
	I	II	III	IV																
	%	%	%	%	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
IPAH																				
...without comorbidities	0	9	72	19	241	151	29.4	13.3	51.2	12.7	9.8	4.6	9.8	5.7	11.8	5.9	62.3	9.9	2.2	0.8
...with comorbidities	0	1	71	28	191	142	32.8	11.5	50.5	9.7	12.1	5.1	12.2	6.0	10.4	4.8	59.8	8.5	2.2	0.7
Connective tissue disease APAH																				
...scleroderma	0	6	81	12	220	142	30.5	12.6	41.5	11.5	10.0	3.9	8.8	5.6	8.8	5.4	64.7	9.6	2.5	0.8
...excluding scleroderma	1	6	78	16	215	141	33.4	11.7	44.0	10.9	10.0	4.8	9.0	5.8	9.1	5.0	63.0	10.1	2.4	0.8
Congenital heart disease APAH	2	14	78	6	277	136	26.3	12.9	44.8	15.4	12.6	5.8	9.4	5.1	6.6	4.7	74.9	10.5	3.1	1.2
Portal hypertension	0	10	78	12	275	157	29.7	12.4	47.7	11.4	11.0	5.1	9.5	6.7	7.7	4.1	67.7	8.9	3.0	1.2

Notes: 1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH. 2. Patients with connective tissue disease where scleroderma is not specified are not included in the table. 3. PH characteristics derived from the closest measurement within 90 days of diagnosis. See the [Glossary: PH Characteristics](#) slides 1 and 2 for further information. 4. SD = Standard deviation. 5. **Parameters** (top row of chart) for some characteristics include normal ranges (e.g. 0 to 7 for RAP), which show the typical range for a healthy person. 6. See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Appendices: PAH type

Lung function tests

Table A 3: Lung function tests at diagnosis by PAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Diagnostic group	Mean per cent predicted forced expiratory volume in one second (FEV1) (%)		Mean per cent predicted forced vital capacity (FVC) (%)		Mean per cent predicted transfer factor for carbon monoxide (TLco) (%)	
	Mean	SD	Mean	SD	Mean	SD
IPAH						
...without comorbidities	85	19	92	20	55	21
...with comorbidities	76	22	88	24	36	18
Connective tissue disease APAH						
...scleroderma	79	31	85	26	37	16
...excluding scleroderma	69	22	75	24	35	16
Congenital heart disease APAH	68	21	79	22	69	26
Portal hypertension	81	18	93	20	50	15

Notes:

- 1. IPAH with comorbidities** is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
- Patients with connective tissue disease where scleroderma is not specified are not included in the table.
- Lung function test results derived from the closest measurement within 90 days of diagnosis. See the [Glossary: Lung function tests](#) slide for further details.
- SD = Standard deviation.
- See the [Survival analysis: Overview](#) slide for details on which patients are included.



Appendices: PAH type

Therapy type

[Glossary](#)

Table A 4: Latest recorded therapy type by PAH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Diagnostic group	Patients	Mono-therapy		Dual therapy		Triple+ therapy		No therapy	
	Number	Patients	%	Patients	%	Patients	%	Patients	%
IPAH									
...without comorbidities	1,132	391	35	580	51	120	11	41	4
...with comorbidities	385	214	56	160	42	5	1	6	2
Connective tissue disease APAH									
...scleroderma	1,045	413	40	561	54	29	3	42	4
...excluding scleroderma	381	185	49	157	41	20	5	19	5
Congenital heart disease APAH	1,117	585	52	264	24	20	2	248	22
Portal hypertension	278	164	59	88	32	2	1	24	9

Notes:

- IPAH with comorbidities** is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
- Patients with connective tissue disease where scleroderma is not specified are not included in the table.
- Therapy type is derived from the number of concurrent drug classes prescribed on the latest recorded therapy date. See the [Glossary: PH Characteristics \(1\)](#) slide for details of how drug class is defined.
- See the [Survival analysis: Overview](#) slide for details on which patients are included.



Appendices: PH type Demographics

[Glossary](#)

Table A 5: Demographics by PH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Diagnostic group	Patients	Age at diagnosis				Female	Days after diagnosis on which the cohort had 50 per cent mortality
	Number	Lower quartile	Median	Upper quartile	Mean	%	
PAH	4,812	46	61	71	58	68	5 years and 53 days
Left Heart Disease	3,265	67	74	79	72	65	4 years and 183 days
Lung Disease	2,093	60	68	75	67	50	1 years and 295 days
CTEPH							
...Operated	1,237	49	61	70	58	45	<i>Not reached at 9 years and 73 days</i>
...Not operated	1,583	58	69	78	66	54	5 years and 272 days
Miscellaneous/uncertain causes	528	52	62	71	61	55	3 years and 96 days

Notes:

1. The **PAH** group includes patients with congenital heart disease APAH, connective tissue disease APAH, IPAH, portal hypertension APAH and other PAH sub-groups.
2. See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Appendices: PH type

PH characteristics

[Glossary](#)

Table A 6: PH characteristics at diagnosis by PH type,

Adult patients in the longitudinal cohort, Great Britain, 2009-19

Parameters	I = least severe IV = most severe				Further = healthier		0=best QoL 50=worst		Norm~14. 25+ =PH		Norm ~8		Norm= 0 to 7		Norm= ≤3		Norm= 65 to 75		Norm= 2.5 to 4.2	
Diagnostic group	WHO functional class				Six-minute walk test (metres)		Quality of life: emPHasis-10 score (0-50)		Mean pulmonary artery pressure (PAP) (mm Hg)		Mean pulmonary wedge pressure (PWP) (mm Hg)		Mean right atrial pressure (RAP) (mm Hg)		Mean pulmonary vascular resistance (PVR) (Wood units)		Mixed venous oxygen saturation (SvO2) (%)		Cardiac index (L/min/m ²)	
	I	II	III	IV																
	%	%	%	%	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
PAH	1	9	76	14	244	145	29.8	12.9	46.3	12.8	10.5	4.8	9.6	5.7	9.6	5.6	64.6	10.4	2.5	0.9
Left Heart Disease	2	12	77	9	211	136	28.4	13	38.8	10.5	21.1	6.0	13.9	5.8	4.2	3.1	64.9	9.4	2.6	0.9
Lung Disease	0	6	71	23	190	129	33.0	11.8	40.0	10.7	11.9	4.9	9.3	5.5	6.8	4.2	65.0	8.7	2.7	1.0
CTEPH																				
...Operated	1	17	74	8	293	138	28.2	12.3	44.2	12.1	11.0	4.9	9.7	5.4	8.5	4.8	63.5	8.6	2.3	0.7
...Not operated	2	13	74	11	253	151	28.9	12.9	41.5	12.2	11.0	4.8	9.1	5.4	8.1	5.2	63.4	8.9	2.3	0.7
Miscellaneous/ uncertain causes	1	10	75	15	224	134	31.2	11.3	42.5	11.2	11.6	5.3	9.2	6.0	7.9	5.1	64.1	9.8	2.7	1.2

Notes:

- The **PAH** group includes patients with congenital heart disease APAH, connective tissue disease APAH, IPAH, portal hypertension APAH and other PAH sub-groups.
- PH characteristics derived from the closest measurement within 90 days of diagnosis. See the [Glossary: PH Characteristics](#) slides 1 and 2 for further information.
- SD = Standard deviation.
- Parameters** (top row of chart) for some characteristics include normal ranges (e.g. 0 to 7 for RAP), which show the typical range for a healthy person.
- See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Appendices: PH type

Lung function tests

Table A 7: Lung function tests at diagnosis by PH type,
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Diagnostic group	Mean per cent predicted forced expiratory volume in one second (FEV1) (%)		Mean per cent predicted forced vital capacity (FVC) (%)		Mean per cent predicted transfer factor for carbon monoxide (TLco) (%)	
	Mean	SD	Mean	SD	Mean	SD
PAH	78	24	86	23	48	22
Left Heart Disease	68	27	76	22	56	23
Lung Disease	54	23	71	26	33	21
CTEPH						
...Operated	84	19	95	20	66	17
...Not operated	78	23	90	22	58	20
Miscellaneous/ uncertain causes	60	22	72	24	35	16

Notes:

1. The **PAH** group includes patients with congenital heart disease APAH, connective tissue disease APAH, IPAH, portal hypertension APAH and other PAH sub-groups.
2. Lung function test results derived from the closest measurement within 90 days of diagnosis. See the [Glossary: Lung function tests](#) slide for further details.
3. SD = Standard deviation.
4. See the [Survival analysis: Overview](#) slide for details on which patients are included.



Appendices: PH type

Therapy type

[Glossary](#)

Table A 8: Latest recorded therapy type by PH type,

Adult patients in the longitudinal cohort, Great Britain, 2009-19

Diagnostic group	Patients	Mono-therapy		Dual therapy		Triple+ therapy		No therapy	
	Number	Patients	%	Patients	%	Patients	%	Patients	%
PAH	4,812	2,144	45	1,945	40	210	4	513	11
Left Heart Disease	3,265	103	3	6	0	0	0	3,156	97
Lung Disease	2,093	402	19	52	2	0	0	1,639	78
CTEPH									
...Operated	1,237	746	60	66	5	0	0	425	34
...Not operated	1,583	924	58	182	11	4	0	473	30
Miscellaneous/ uncertain causes	528	196	37	96	18	7	1	229	43

Notes:

1. The **PAH** group includes patients with congenital heart disease APAH, connective tissue disease APAH, IPAH, portal hypertension APAH and other PAH sub-groups.
2. Therapy type is derived from the number of concurrent drug classes prescribed on the latest recorded therapy date. See the [Glossary: PH Characteristics \(1\)](#) slide for details of how drug class is defined.
3. See the [Survival analysis: Overview](#) slide for details on which patients are included.



Appendices: 2019 IPAH review

Summary

Overview: Previous annual reports have suggested that the presence of other diseases of the heart, circulation and respiratory system in patients with IPAH may affect their outcome.

In November 2018 lead clinicians at the PH centres agreed to review the diagnosis of all patients in the NAPH by May 2019. They took all patients with a diagnosis of idiopathic, heritable, and anorexigen-induced pulmonary arterial hypertension (IPAH) and divided them into those with and without other diseases of the heart, circulation and respiratory system which may be contributing to PH.

Objective: The purpose of the review was to get more meaningful survival data for IPAH patients by subdividing patients with IPAH into those with and without significant comorbid diseases.

Definition: The following criteria were used:

- IPAH includes patients who meet the clinical criteria according to the ESC/ERS guidelines¹ and who do not have clinically relevant left heart disease, parenchymal lung disease or hypoxia.
- If comorbid conditions are present, they must be minor and not contributory to PH. The patient must be suitable for parenteral prostacyclin therapy, at least theoretically.
- The term IPAH without comorbidities does not apply to patients in whom left heart disease or parenchymal lung disease or hypoxia is contributing to PH. Patients with any additional PH diagnosis in the NAPH were also excluded.

IPAH codes in NAPH

Main diagnosis		Main diagnosis subcategory
1	Pulmonary Arterial Hypertension	1.1 Idiopathic PAH
		1.2.1 Heritable PAH - BMPR2
		1.2.2 Heritable PAH - ALK-1, endoglin (with or without HHT)
		1.2.3 Heritable PAH - Unknown
		1.3 Drugs and toxins induced



Appendices: 2019 IPAH review

Impact

Table A 9: Changes to IPAH diagnoses following the 2019 IPAH Review,

All patients, Great Britain, 2009-19

Description	Main diagnosis group		
	9AR	10AR	Patients
Main diagnosis remains IPAH, split by comorbidities	IPAH	IPAH without comorbidities	1,657
		IPAH with comorbidities	420
	Total		2,077
Main diagnosis no longer IPAH	IPAH	Lung Disease	215
		Left Heart Disease	29
		Congenital Heart Disease	23
		Connective Tissue Disease	15
		Other diagnoses	39
		Patient deleted	0
	Total		321
New main diagnosis of IPAH	Other diagnoses	IPAH without comorbidities	59
		IPAH with comorbidities	20
	Not yet entered	IPAH without comorbidities	86
		IPAH with comorbidities	20
	Total		185

The NAPH team are grateful to all PH centres for their hard work reviewing patients with IPAH diagnoses. The improved survival curve for IPAH on the following slide suggests that the impact of comorbidity on IPAH survival has been reduced, resulting in more robust outputs.

Key finding

69 per cent of patients with IPAH diagnoses in the 9AR extract were defined as IPAH without comorbidities in the 10AR.

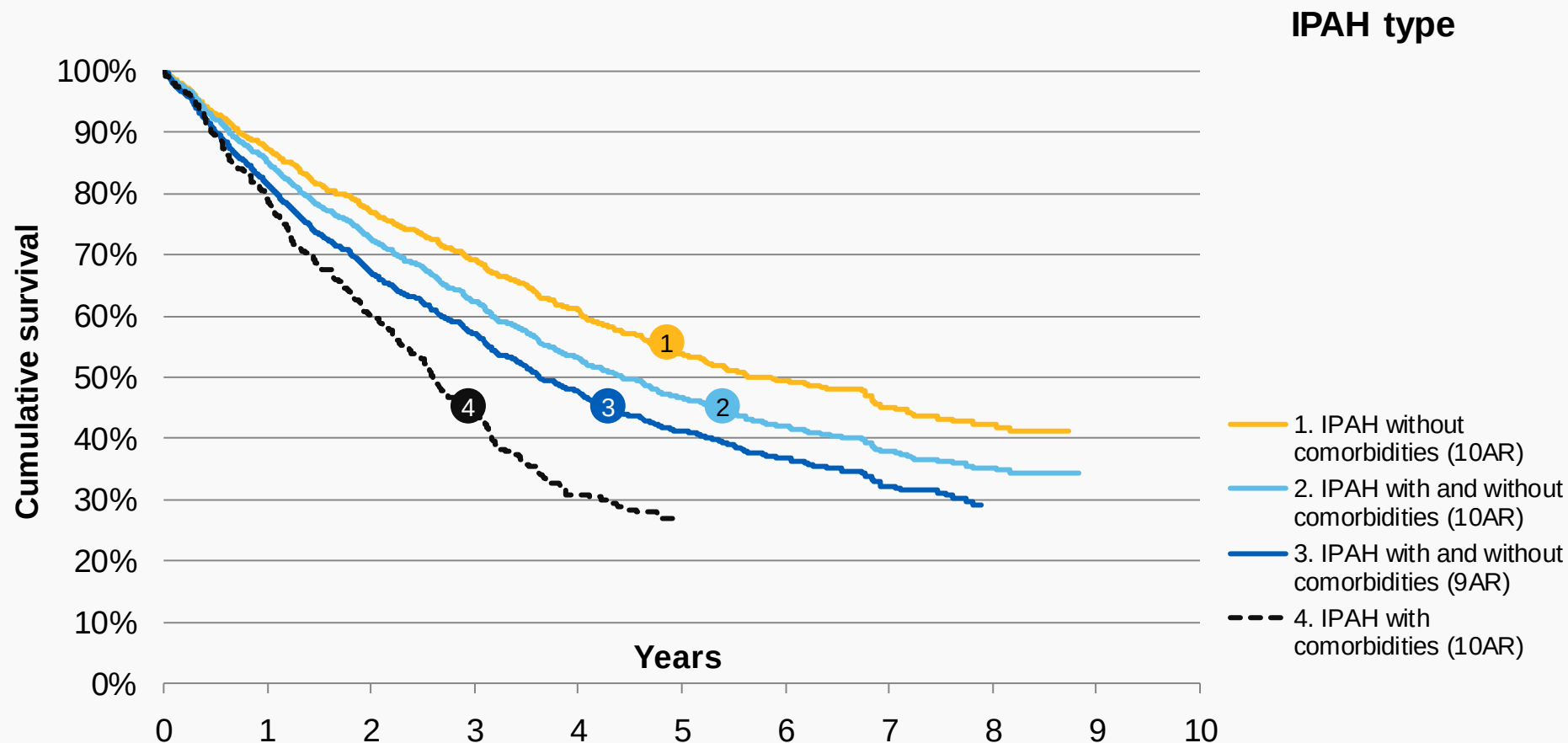
Notes: 1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
2. Table A 9 includes all patients in the NAPH, not just those in the longitudinal cohort. This allows all updates to IPAH patients to be shown.



Appendices: 2019 IPAH review

Survival curves

Chart A 1: Survival curves from first IPAH diagnosis by audit report (AR),
Adult patients in the longitudinal cohort, Great Britain, 2009-19



Notes: 1. **IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.

2. See the [Survival analysis: Overview](#) slide for details on which patients are included.

3. The survival curve is Stopped if the number of patients used to calculate the cumulative survival drops below 50.



Appendices: 2019 IPAH review

Annual survival

Table A 10: Post-diagnosis survival of IPAH patients by audit report (AR),
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Years Elapsed	IPAH							
	9AR		10AR					
	With and without comorbidities		With and without comorbidities		Without comorbidities		With comorbidities	
	Patients	% survival	Patients	% survival	Patients	% survival	Patients	% survival
0	1,645	100	1,517	100	1,132	100	385	100
1	1,178	81	1,165	85	880	87	285	79
2	848	67	864	72	675	77	189	60
3	617	57	653	62	537	69	116	43
4	421	47	474	53	407	61	67	31
5	284	41	336	47	288	54	–	–
6	181	37	237	42	206	49	–	–
7	96	32	151	38	135	45	–	–
8	–	–	85	35	78	42	–	–
9	–	–	–	–	–	–	–	–

Notes:

- IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
- See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Appendices: 2019 IPAH review

Demographics

[Glossary](#)

Table A 11: Demographics of IPAH patients by audit report (AR),
Adult patients in the longitudinal cohort, Great Britain, 2009-19

IPAH type	Patients	Age at diagnosis				Female	Days after diagnosis on which the cohort had 50 per cent mortality
	Number	Lower quartile	Median	Upper quartile	Mean	%	
9AR							
...With and without comorbidities	1,645	50	66	74	62	61	3 years and 227 days
10AR							
...With and without comorbidities	1,517	48	64	73	60	62	4 years and 153 days
...Without comorbidities	1,132	45	60	72	58	65	5 years and 293 days
...With comorbidities	385	63	70	77	68	53	2 years and 218 days

Notes:

- IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
- See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Appendices: 2019 IPAH review

PH characteristics

[Glossary](#)

Table A 12: PH characteristics of IPAH patients at diagnosis by audit report (AR),
Adult patients in the longitudinal cohort, Great Britain, 2009-19

Parameters	I = least severe IV = most severe				Further = healthier		0=best QoL 50=worst		Norm~14. 25+ =PH		Norm ~8		Norm= 0 to 7		Norm= ≤3		Norm= 65 to 75		Norm= 2.5 to 4.2	
IPAH type	WHO functional class				Six-minute walk test (metres)		Quality of life: emPHasis-10 score (0-50)		Mean pulmonary artery pressure (PAP) (mm Hg)		Mean pulmonary wedge pressure (PWP) (mm Hg)		Mean right atrial pressure (RAP) (mm Hg)		Mean pulmonary vascular resistance (PVR) (Wood units)		Mixed venous oxygen saturation (SvO2) (%)		Cardiac index (L/min/m ²)	
	I	II	III	IV																
	%	%	%	%	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
9AR																				
...With and without comorbidities	0	6	71	23	230	149	30.4	12.6	50.5	11.9	10.8	5.2	10.7	6.1	11.1	5.5	61.5	9.5	2.2	0.7
10AR																				
...With and without comorbidities	0	7	72	21	236	151	30.2	13.0	51.0	12.0	10.4	4.8	10.4	5.9	11.5	5.7	61.7	9.6	2.2	0.7
...Without comorbidities	0	9	72	19	241	151	29.4	13.3	51.2	12.7	9.8	4.6	9.8	5.7	11.8	5.9	62.3	9.9	2.2	0.8
...With comorbidities	0	1	71	28	191	142	32.8	11.5	50.5	9.7	12.1	5.1	12.2	6.0	10.4	4.8	59.8	8.5	2.2	0.7

Notes:

- 1. IPAH without comorbidities** excludes the following: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
- PH characteristics derived from the closest measurement within 90 days of diagnosis. See the [Glossary: PH Characteristics](#) slides 1 and 2 for further information.
- SD = Standard deviation.
- Parameters** (top row of chart) for some characteristics include normal ranges (e.g. 0 to 7 for RAP), which show the typical range for a healthy person.
- See the [Survival analysis: Overview](#) slide for details on which patients are included and when the survival curve is stopped.



Appendices: 2019 IPAH review

Lung function tests

[Glossary](#)

Table A 13: Lung function tests of IPAH patients at diagnosis by audit report (AR),
Adult patients in the longitudinal cohort, Great Britain, 2009-19

IPAH type	Mean per cent predicted forced expiratory volume in one second (FEV1) (%)		Mean per cent predicted forced vital capacity (FVC) (%)		Mean per cent predicted transfer factor for carbon monoxide (TLco) (%)	
	Mean	SD	Mean	SD	Mean	SD
9AR						
...With and without comorbidities	80	22	90	22	47	22
10AR						
...Without comorbidities	83	20	91	21	51	22
...With comorbidities	85	19	92	20	55	21
...With and without comorbidities	76	22	88	24	36	18

Notes:

- IPAH with comorbidities** is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
- Lung function test results derived from the closest measurement within 90 days of diagnosis. See the [Glossary: Lung function tests](#) slide for further details.
- SD = Standard deviation.
- See the [Survival analysis: Overview](#) slide for details on which patients are included.



Appendices: 2019 IPAH review

Therapy type

[Glossary](#)

Table A 14: Latest recorded therapy type of IPAH patients by audit report (AR),
Adult patients in the longitudinal cohort, Great Britain, 2009-19

IPAH type	Patients	Mono-therapy		Dual therapy		Triple+ therapy		No therapy	
	Number	Patients	%	Patients	%	Patients	%	Patients	%
9AR									
...With and without comorbidities	1,645	788	48	706	43	86	5	65	4
10AR									
...With and without comorbidities	1,517	605	40	740	49	125	8	47	3
...Without comorbidities	1,132	391	35	580	51	120	11	41	4
...With comorbidities	385	214	56	160	42	5	1	6	2

Notes:

- IPAH with comorbidities** is defined as: IPAH in which an additional clinically significant contribution to PH may be present from other cardiac or respiratory disease(s). This has been indicated in the NAPH by recording an additional cause of PH.
- Therapy type is derived from the number of concurrent drug classes prescribed on the latest recorded therapy date. See the [Glossary: PH Characteristics \(1\)](#) slide for details of how drug class is defined.
- See the [Survival analysis: Overview](#) slide for details on which patients are included.



National Audit of Pulmonary Hypertension

Methodology

National Standards
and changes



Methodology: Changes to National Standards

National Standard		Change		
No.	Description	9 th Annual Report	10 th Annual Report	Reason
2	PH centres should see a sufficient number of patients	250 PAH / CTEPH adult patients	300 PAH / CTEPH adult patients	Target numbers increased to reflect the national increase in the number of patients managed by the specialist PH centres.
		50 PAH / CTEPH child patients	100 PAH / CTEPH child patients	
4	New patients should be seen or discharged within 30 days	95% of all adult patients (no change)		This standard has been altered in the children's PH centre to focus on patients with PAH and CTEPH because these patients may require urgent treatment which is only available at a specialist PH centre.
		95% of all child patients	95% of PAH / CTEPH child patients	
5	Patients receiving a PH drug should have pre-treatment cardiac catheterization	Cardiac catheterization before first drug	Cardiac catheterization within 2 weeks of first drug	To account for patients who need to be stabilized with prostacyclin before cardiac catheterization.
7	Patients should have a pre-treatment vasoreactivity study recorded	N/A	New Target = 80%.	New standard, reported nationally for the 10 th Annual Report and at centre-level in subsequent reports. The information provided by undergoing a vasoreactivity study determines whether the patient with IPAH is most likely to respond to treatment with calcium channel blockers or other drug therapies including prostanoids, endothelin antagonists or PDE5 inhibitors.
12	Patients receiving a PH drug should have an annual consultation	Target = 95%	Target = 90%	To take account off more virtual consultations for sick patients unable to travel to their designated PH centre.



Methodology: National Standards 1 to 3

National Standard		Methodology	
No.	Description	Denominator Cases assessed	Numerator How met?
1	PH centres should participate in the Audit [<i>all PH centres</i>]	All PH centres that treat patients in England, Scotland and Wales.	PH centres will determine if they are participating. Failure to participate will remove them from all other standards.
2a	Adult PH centres should see a sufficient number of patients with PAH or CTEPH [<i>adult PH centres</i>]	All PH centres that treat adult patients in England, Scotland and Wales.	PH centres are expected to manage ¹ at least 300 adult patients per annum with pulmonary arterial hypertension (PAH) or chronic thromboembolic pulmonary hypertension (CTEPH).
2b	The children's PH centre should see a sufficient number of patients [<i>Great Ormond Street only</i>]	The specialist children's PH centre that treats patients from England, Scotland and Wales.	The specialist centre is expected to manage ¹ at least 300 patients per annum, of whom 100 patients per annum with pulmonary arterial hypertension (PAH) or chronic thromboembolic pulmonary hypertension (CTEPH).
3	Patients should have a timely diagnosis [<i>adults and children</i>]	Patients with a first referral letter received between 6 months before the audit start date and 6 months before the audit end date.	Patients in the denominator with a diagnosis recorded.



Methodology: National Standards 4 to 6

National Standard		Methodology	
No.	Description	Denominator Cases assessed	Numerator How met?
4a	New patients should be seen or discharged within 30 days [adults only]	Patients with a first referral letter received between 30 days before the audit start date and 30 days before the audit end date.	Patients in the denominator with a consultation (inpatient, day case or outpatient) or discharge within 30 days following the date of the first referral.
4b	New PAH/CTEPH patients should be seen or discharged within 30 days [children only]	Patients with a latest diagnosis of PAH or CTEPH, where a first referral letter was received between 30 days before the audit start date and 30 days before the audit end date.	Patients in the denominator with a consultation (inpatient, day case or outpatient) or discharge within 30 days following the date of the first referral.
5	Patients receiving a PH drug should have pre-treatment cardiac catheterization [adults only]	Patients with a first drug in the audit year (whose latest diagnosis is not congenital heart disease and who did not start their drug before referral).	Patients in the denominator with a cardiac catheterization that took place either before, or within 14 days, of the date of the first drug.
6	Patients should have pre-treatment WHO functional class and an exercise test recorded [adults only]	Patients with a latest diagnosis of PAH or CTEPH, whose first pulmonary hypertension drug is in the audit period and who did not start a pulmonary hypertension drug before their first referral.	Patients in the denominator who have a WHO functional class and an exercise test recorded before the starting date of their first pulmonary hypertension drug.



Methodology: National Standards 7 to 9

National Standard		Methodology	
No.	Description	Denominator Cases assessed	Numerator How met?
7	Patients should have a pre-treatment vasoreactivity study recorded [<i>adults only</i>]	Patients with a latest diagnosis of idiopathic, heritable or drug-induced PAH whose first pulmonary hypertension drug is in the audit period and who did not start a pulmonary hypertension drug before their first referral.	Patients in the denominator with a vasoreactivity test involving nitric oxide, prostanoïd or adenosine recorded before the starting date of their first pulmonary hypertension drug.
8	New patients should begin drug therapy within 12 weeks of referral [<i>adults only</i>]	Patients with a first referral letter received between 12 weeks before the audit start date and 12 weeks before the audit end date, and a latest diagnosis of idiopathic, heritable, anorexigen-induced or connective tissue disease associated PAH, and with a highest WHO functional class of II, III, or IV who had not started any first drug therapy before their first referral date.	Patients in the denominator with a first drug therapy within 12 weeks of first referral.
9	Patients receiving a PH drug should have had a PH diagnosis recorded [<i>adults only</i>]	Patients with an active drug therapy at the end of the audit year, who did not start a drug before their first referral.	Patients in the denominator with a latest diagnosis of pulmonary hypertension.



Methodology: National Standards 10 to 13

National Standard		Methodology	
No.	Description	Denominator Cases assessed	Numerator How met?
10	First line drug therapy for PAH should be a phosphodiesterase 5 (PDE5) inhibitor [<i>adults only</i>]	Patients who started their first drug (where the first drug was not a calcium channel blocker) between the audit start date and audit end date, whose latest diagnosis was Pulmonary Arterial Hypertension.	Patients in the denominator whose first drug was exclusively a phosphodiesterase 5 (PDE5) inhibitor.
11	Patient quality of life should be recorded [<i>adults only</i>]	Patients with a latest diagnosis of PAH or CTEPH with a consultation (inpatient, day case or outpatient) between the audit start date and audit end date.	Patients in the denominator that have at least one quality of life questionnaire recorded between the audit start date and audit end date.
12	Patients receiving a PH drug should have an annual consultation [<i>adults and children</i>]	Patients with an active drug therapy at the end of the audit year, who did not start a drug before their first referral.	Patients in the denominator who had a consultation between 1 month before the audit start date and audit end date.
13	Waiting times for pulmonary endarterectomy should be <4 months [<i>adults only</i>]	Patients having their first pulmonary endarterectomy in the audit year with a first diagnosis of CTEPH before the date of pulmonary endarterectomy.	Patients in the denominator where first diagnosis of CTEPH is ≤ 4 months before pulmonary endarterectomy.



National Audit of Pulmonary Hypertension

Glossary



Glossary: Specialist PH centres

Patients with pulmonary hypertension are treated at one of eight specialist pulmonary hypertension centres in the UK.

Special Health Board / NHS trust of PH centre		Known as
Golden Jubilee National Hospital	Glasgow	Golden Jubilee
Great Ormond Street Hospital for Children NHS Foundation Trust ¹	London	Great Ormond Street
Imperial College Healthcare NHS Trust	London	Imperial College
The Newcastle Upon Tyne Hospitals NHS Foundation Trust	Newcastle	Newcastle
Royal Brompton and Harefield NHS Foundation Trust	London	Royal Brompton and Harefield
Royal Free London NHS Foundation Trust	London	Royal Free
Royal Papworth Hospital NHS Foundation Trust ²	Cambridgeshire	Royal Papworth
Sheffield Teaching Hospitals NHS Foundation Trust	Sheffield	Sheffield

Notes:

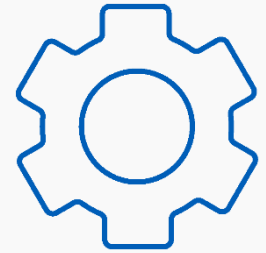
1. Great Ormond Street provides care for children with pulmonary hypertension.
2. Royal Papworth Hospital is the only surgical centre in the UK for pulmonary endarterectomy.



Glossary: Patient cohorts used in the report

The following patient cohorts are referred to in the report:

- **Managed patients** have a referral active at any point during the appropriate year.
- **Active patients** have a referral active on the 31 March of the appropriate year.
- **Active drug therapies** represent patients with at least one pulmonary hypertension drug therapy being prescribed on the 31 March of the appropriate year.
- The **cohort for longitudinal analysis** are patients whose first referral was on or after 1 April 2009. This cohort is used when there is a need to analyse data where the full treatment history for a patient is required. Since the Audit started on 1 April 2009, it is reasonable to expect that all of the events after that date will be recorded.



Scottish patients are submitted to the Audit by Golden Jubilee National Hospital using a pseudonymised patient identifier.

Therefore NHS Digital cannot identify a Scottish patient, although Golden Jubilee can. As a consequence:

- a) It is not possible for NHS Digital to identify patients who have been treated in both Scotland and England. This means that a small number of patients may be included twice in the cohort for longitudinal analysis, once for the English recorded activity, and once for the Scottish.
- b) It is not possible for the Office for National Statistics (ONS) to identify mortality data for Scottish patients. For English and Welsh patients, around one fifth of deaths are identified by ONS, not the Audit. It is therefore likely that some Scottish deaths will be missed by the Audit, though the proportion missing is not known.



Glossary: Acronyms and definitions (1)

Cardiac catheterization

An invasive diagnostic procedure that provides information about the structure and function of the heart. Cardiac catheterization is considered the gold standard method for confirming the presence, nature and severity of pulmonary hypertension.

CTEPH Chronic thromboembolic pulmonary hypertension

In chronic thromboembolic pulmonary hypertension (WHO Group 4) blood vessels are blocked or narrowed by the chronic consequences of blood clots, causing raised blood pressure in the pulmonary arteries.

NAPH National Audit of Pulmonary Hypertension

The National Audit of Pulmonary Hypertension measures the quality of care provided to people referred to pulmonary hypertension services in Great Britain.

PAH Pulmonary arterial hypertension

(APAH) In pulmonary arterial hypertension (WHO Group 1) the blood vessels become narrowed as the arterial walls stiffen and thicken, causing raised blood pressure in the pulmonary arteries. PAH resulting from another disease or cause such as connective tissues disease or congenital heart diseases is called associated-PAH or APAH.

PEA Pulmonary endarterectomy

This is a complex surgical procedure undertaken in selected patients with CTEPH, involving a heart-lung bypass and induced hypothermia where the circulation is slowed or stopped while blood clots and scar tissue are removed from the pulmonary arteries. The procedure can restore the raised blood pressure in the pulmonary circulation to normal in some patients with CTEPH.



Glossary: Acronyms and definitions (2)

PH Pulmonary hypertension

This is a medical condition in which the blood pressure is elevated in the blood vessels which supply the lungs. It can damage the heart and reduce its efficiency in supplying oxygen around the body.

PHA UK Pulmonary Hypertension Association UK

The only UK charity dedicated to people with pulmonary hypertension – it supports patients, works with healthcare professionals, provides research grants and lobbies for pulmonary hypertension.

phosphodiesterase 5 (PDE5) inhibitors

Phosphodiesterase 5 (PDE5) inhibitors are a type of targeted therapy used to treat some people with pulmonary hypertension. PDE5 inhibitors (i.e. sildenafil and tadalafil) cause the blood vessels to relax by stopping the PDE5 enzyme from working properly. This increases blood flow to the lungs and lowers blood pressure.

Significance

National results have been compared to those in the previous report to test whether changes over time are statistically significant at the 95% level ($p \leq 0.05$). This means that the result of the significance test has a 95% chance of being true¹ ([see Glossary: Significance testing](#)).

WHO World Health Organization

The World Health Organization is the United Nations coordinating authority on international health.



Glossary: PH characteristics (1)

The **six-minute walking test** involves a person walking as far as possible up and down a corridor in six minutes. During the test, blood pressure, heart rate and oxygen levels are also measured. The test can provide clues to the cause of some symptoms of PH and is a good way to measure how effective PH treatment is, and if any changes to treatment are needed.



emPHasis-10 is a questionnaire which measures the impact of PH on a person's life, and how this changes over time. A total emPHasis-10 score is derived using simple aggregation of the 10 items. Scores range from zero to 50, with higher scores indicating worse quality of life as a result of the condition.

A patient's PH drug **therapy type** is defined as either:

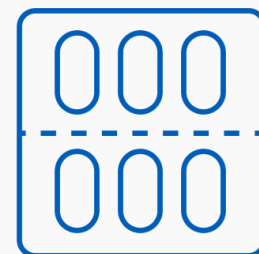
- Mono-therapy
- Dual therapy
- Triple+ therapy
- No therapy



The therapy type is derived from the number of concurrent drug classes prescribed on the latest recorded therapy date.

The four **drug classes** are:

- 1) Phosphodiesterase 5 inhibitors;
- 2) Endothelin receptor antagonists;
- 3) Prostanoids;
- 4) Calcium channel blockers for vasoreactive PAH3.



Triple+ therapy also covers a small number of patients receiving all four drug classes concurrently.



Glossary: PH characteristics (2)

Mean pulmonary artery pressure (PAP) at rest is measured by cardiac catheterization. The normal mean pressure is 14 mm Hg and a person is considered to have PH when they have a mean baseline greater than or equal to 25 mm Hg¹.

Mean pulmonary wedge pressure (PWP) is an estimation of blood pressure on the left side of the heart (blood returning from the lungs to the heart), as measured by cardiac catheterization. The normal mean pressure is 8 mm Hg.

Mean right atrial pressure (RAP) is the blood pressure in the right atrium of the heart, as measured by cardiac catheterization, and is known to be affected by PH. A normal range is 0 to 7 mm Hg¹.



Mean pulmonary vascular resistance (PVR) is a measure of the resistance that must be overcome to push blood through the pulmonary circulatory system. A normal range is less than 3 Wood Units.

SvO₂ (also known as mixed venous oxygen saturation) is the percentage of oxygen saturation in the pulmonary arterial blood, which is a measure of how much blood the heart is pumping round the body. The normal range is 65 to 75 per cent but it needs to be interpreted in the context of the clinical problem.

Cardiac index is a haemodynamic parameter that assesses how much blood a person's heart is pumping around the body. The cardiac index takes into account body size by dividing a person's cardiac output by their body surface area. The normal range for cardiac index is 2.5 to 4.2 L/min/m².



Glossary: Lung function tests

A variety of lung function tests are used to help diagnose and monitor lung conditions such as PH.

Mean forced expiratory volume in one second (FEV1) measures how much air a person can exhale during the first second of a forced breath.

Mean forced vital capacity (FVC) measures how much air a person can exhale in one complete forced breath after taking a deep breath in.

Mean transfer factor for carbon monoxide (TLco) measures how well a person's lungs are able to take up oxygen from the air they breathe and transfer it into the blood.

To become meaningful, results from lung function tests need to be adjusted for the patient's size and demographics. The equations in the table below use sex, age and height to produce predicted values for each patient. The measured results from each lung function test can then be expressed as a **per cent predicted** e.g. 50 per cent if the measured result is half the predicted result.

Equations to produce lung function test predictions¹

Lung function test	Equation	
	Female A = 18 to 70 H = 1.45 to 1.80	Male A = 18 to 70 H = 1.55 to 1.95
FEV1	$3.95H - 0.025A - 2.60$	$4.30H - 0.029A - 2.49$
FVC	$4.43H - 0.026A - 2.89$	$5.76H - 0.026A - 4.34$
TLco	$8.18H - 0.049A - 2.74$	$11.11H - 0.066A - 6.03$

Key: A = Age in years. H = Height in metres



Glossary: Significance testing

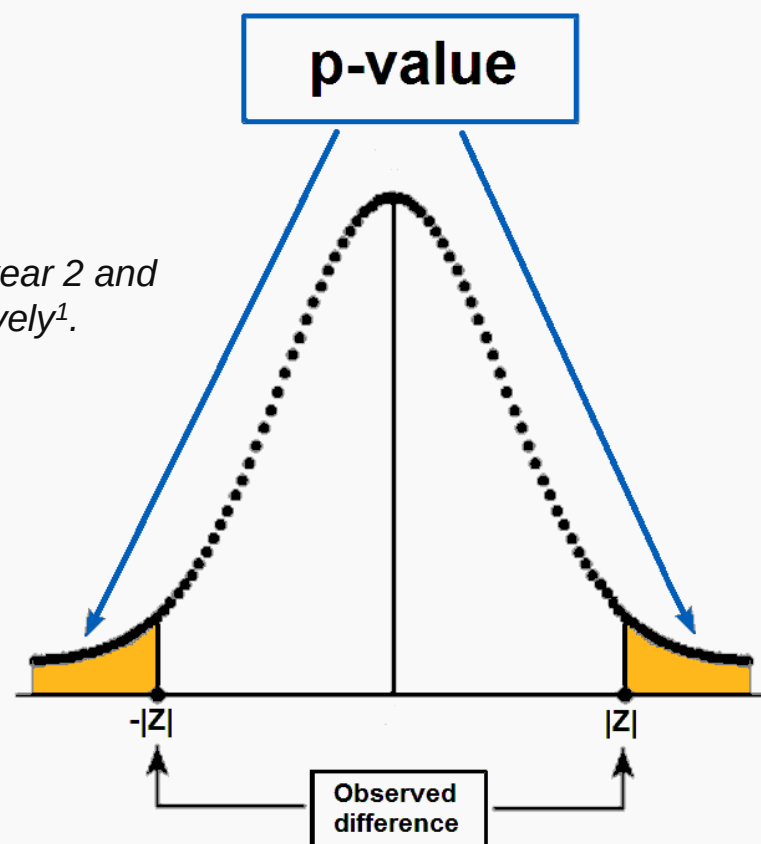
In the report, national results have been compared to those in the previous report to test whether changes over time are **statistically significant** at the 95% level ($p \leq 0.05$).

First the **standard error** (SE) of the difference between the results from the two audit years is calculated using the following formula:

$$SE(p_1 - p_2) = \sqrt{\left(\frac{p_1(100 - p_1)}{n_1} + \frac{p_2(100 - p_2)}{n_2} \right)}$$

Where p_1 is the percentage in year 1 and p_2 is percentage in year 2 and where n_1 and n_2 are the sample sizes in year 1 and 2 respectively¹.

The SE is then used to plot a normal distribution curve for the combined years. The **observed difference** in percentage points between years 1 and 2 is plotted on the x axis and used to derive the **p value**, the calculated probability of the observed difference being consistent with the **null hypothesis** (no difference between the audit years). A p value of ≤ 0.05 means that there is a 95% chance that the observed difference reflects a real difference between audit years and is not the result of normal variation.



Glossary: World Health Organization (WHO)

Functional Class

The severity of pulmonary hypertension symptoms are recorded using the World Health Organization (WHO) functional class:

- **Class I:** Patients with pulmonary hypertension but without resulting limitation of physical activity. Ordinary physical activity does not cause undue dyspnoea or fatigue, chest pain or near syncope.
- **Class II:** Patients with pulmonary hypertension resulting in slight limitation of physical activity. They are comfortable at rest. Ordinary physical activity causes undue dyspnoea or fatigue, chest pain or near syncope.
- **Class III:** Patients with pulmonary hypertension resulting in marked limitation of physical activity. They are comfortable at rest. Less than ordinary activity causes undue dyspnoea or fatigue, chest pain or near syncope.
- **Class IV:** Patients with pulmonary hypertension with inability to carry out any physical activity without symptoms. These patients manifest signs of right heart failure. Dyspnoea and/or fatigue may even be present at rest. Discomfort is increased by any physical activity.



This is akin to New York Heart Association functional class. Efforts are made to record a WHO functional class for all patients at each hospital visit but this is not possible in all cases.



Glossary: Waiting times for PEA should be <4 months: Background

National Standard 13: Waiting times for pulmonary endarterectomy should be less than 4 months

The Audit can accurately measure the time from chronic thromboembolic pulmonary hypertension (CTEPH) diagnosis to surgery. The date of diagnosis is the date of the multidisciplinary team (MDT) meeting at the specialist PH centre when the diagnosis of CTEPH was confirmed.

The time period of 4 months was chosen by the commissioners in 2016, and is based on the patient entering the 18 week pathway as described in the NHS England PH service specification [A11/S/1](#) 2013-14:

“on receipt of all information on appropriate investigations, all patients who are eligible for pulmonary endarterectomy (PEA) and who would consider surgery, will receive surgery within 18 weeks (in line with the national referral to treatment pathway). However, the average waiting time will be less than 3 months and patients with a deteriorating condition will need an earlier operation. For Papworth patients the 18 week pathway starts once the patient has been assessed as suitable for surgery. For patients referred from external PH centres the PEA 18 week pathway starts on the MDT acceptance of suitability for listing. The clock stops once the patient has been admitted for PEA surgery.” (p.5).

The NAPH 10th Annual Report for 2018-19 reported a median time from diagnosis to surgery of 26 weeks (29 weeks in 2017-18) – see Chart R 2 on [Slide 61](#). Since the patient pathway is complex it may not be feasible to achieve 18 weeks after it is agreed that the patient is operable.



National Audit of Pulmonary Hypertension

Further information



Further information: Explanatory notes

Explanatory note for Golden Jubilee: Data submission and reporting

Scottish data is submitted to the Audit using a pseudonymised patient identifier (see [Slide 89](#) for further information).

Due to different information governance rules in Scotland, data submitted from Golden Jubilee is anonymised and only the first part of the postcodes and the month and year of birth/death are provided. For reporting, Golden Jubilee dates of birth are considered to be the first day of the provided month, while their dates of death are considered to be the last day of the provided month. For English and Welsh patients, the exact dates of birth and death are used.

Scotland are not included in the lung disease or left heart disease survival curves due to difficulties in obtaining death data for those patients (see [Slide 41](#) for further information).

Explanatory note for Newcastle: Data submission and reporting

Newcastle operates a shared care clinic in Hull and East Yorkshire Hospitals NHS Trust (RWA). Hull submits their own data to the Audit. For reporting, the Hull data is combined with the Newcastle data and reported as 'Newcastle'.

Explanatory note for Imperial College: National Standard 5: Data submission

Data submitted from Imperial contained inaccuracies in recording of the date of catheterisation and / or therapy start date in four cases. Imperial will update the data for future annual reports.

Additional explanatory notes are provided in the accompanying **Data Quality Statement**.



Further information: Members of the National Audit of Pulmonary Hypertension Project Board

Name	Role	Organisation
Simon Gibbs (Chair)	Clinical Lead and Chief Auditor (Project executive)	Imperial College London and Imperial College Healthcare NHS Trust
David Kiely	Senior User (Physician)	Chair, Pulmonary Hypertension CRG, NHS England
Kathy Blacker	Senior User (Commissioners)	Regional Programme of Care Manager – Internal Medicine (North), NHS England
Jenny Thomas	Senior User (Commissioners)	Medical Director, Welsh Health Specialised Services Committee
Craig Wheelans	Senior User (Commissioners)	National Medical Advisor, NHS National Services Scotland
Veronica Gillen	Senior User (Commissioners)	Commissioning Lead for Specialist Services, The Health and Social Care Board, Northern Ireland
Paul Corris	Senior User (Physician)	Service Review of PH Centres
Gerry Coghlan	Senior User (Physician)	Chair of PH Physicians Committee
Natalie Doughty	Senior User (Nursing)	Royal Papworth Hospital NHS Foundation Trust
Martin Johnson	Senior User (Physician)	NHS Greater Glasgow & Clyde
Shahin Moledina	Senior User (Physician)	Paediatric Pulmonary Hypertension Service
Kay Yeowart	Senior User (Patients)	PHA UK
Iain Armstrong	Senior User (Patients)	PHA UK
Julie Michalowski	Project Manager	NHS Digital
Chris Dew	Clinical Audit Programme Head	NHS Digital



A background image of a dense bundle of fiber optic cables, with light reflecting off the strands, creating a blue and white bokeh effect.

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